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A generalist foundation model and database for open-world medical image segmentation

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Vision foundation models have demonstrated vast potential in achieving generalist medical segmentation capability, providing a versatile, task-agnostic solution through a single model. However, current generalist models involve simple pre-training on various medical data containing irrelevant information, often resulting in the negative transfer phenomenon and degenerated performance. Furthermore, the practical applicability of foundation models across diverse open-world scenarios, especially in out-of-distribution (OOD) settings, has not been extensively evaluated. Here we construct a publicly accessible database, MedSegDB, based on a tree-structured hierarchy and annotated from 129 public medical segmentation repositories and 5 in-house datasets. We further propose a Generalist Medical Segmentation model (MedSegX), a vision foundation model trained with a model-agnostic Contextual Mixture of Adapter Experts (ConMoAE) for open-world segmentation. We conduct a comprehensive evaluation of MedSegX across a range of medical segmentation tasks. Experimental results indicate that MedSegX achieves state-of-the-art performance across various modalities and organ systems in in-distribution (ID) settings. In OOD and real-world clinical settings, MedSegX consistently maintains its performance in both zero-shot and data-efficient generalization, outperforming other foundation models.

Segmentation is a fundamental process in medical imaging that involves the accurate identification and delineation of regions of interest, such as organs or lesions, from raw medical image data[1]. As such, it plays a crucial role in the radiological workflow by assisting radiologists in conducting precise medical imaging analysis. In addition, segmentation contributes to enhancing disease diagnosis, treatment decision-making, surgical assistance and prognosis assessment, thus optimizing patient management and medical outcomes[2–7]. In recent years, artificial intelligence (AI)[8–12] has shown potential in medical segmentation, obtaining outcomes matching or even surpassing those of clinical experts. Traditional AI-based segmentation approaches, including U-Net[11] and its variants[8–10,12], have shown impressive performance in medical segmentation; however, these segmentation approaches are primarily designed for specific tasks or modalities, making it challenging to generalize to new tasks that were not encountered during model training. While approaches such as DoDNet[13] and UniSeg[14] attempt to transform the task-specific segmentation paradigm into a general one, their clinical adaptability is hindered by the small model parameter size and limited processing capabilities for complex medical imaging tasks.

The rapid advancement of vision foundation models has demonstrated potential in building generalist models by modelling multiple

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tasks into a unified formulation through large-scale pre-training on various datasets[15,16]. There has been an increasing interest in applying these foundation models to medical image segmentation[17], with the aim of achieving universal segmentation that enables versatile downstream tasks without additional models. For example, MedSAM[18] and SAM-Med2D[19] have explored the adaptation of the latest natural image segmentation foundation model, Segment Anything Model (SAM)[15], to segment diverse anatomical structures and lesions across various medical imaging modalities with a single model.

While vision foundation models for generalist segmentation represent a promising research direction, they remain at a preliminary stage towards open-world medical segmentation. Existing research[18,19] employs simple strategies that directly apply foundation models (for example, SAM) or fine-tune diverse medical images aggregated from multiple sources with limited organization. However, for a specific downstream task, involving irrelevant data in pre-training might degenerate the downstream performance[20], a phenomenon known as negative transfer[21]. In addition, the unified modelling to process diverse downstream tasks using shared parameters would face the 'task conflicts' issue[22], thus degrading the model's capacity and hindering its ability to consistently perform well across unseen scenarios. Furthermore, existing methods tend to evaluate their models in in-distribution (ID) settings, while their practical applicability in the open-world setting, especially in addressing out-of-distribution (OOD) shifts between source and target data, has not been fully investigated yet. This highlights the need for the development and comprehensive evaluation of domain-adaptive generalist techniques to scale well for various downstream tasks.

To address these challenges, we introduce an enhanced Generalist Medical Segmentation model (MedSegX), a vision foundation model that introduces Contextual Mixture of Adapter Experts (ConMoAE) for open-world medical segmentation. First, to develop and validate MedSegX, we introduce MedSegDB, an extensive and well-organized medical segmentation database. MedSegDB contains 1,672,275 image–mask pairs from 134 datasets, covering 10 medical imaging modalities, 39 major organs and tissues, and 111 segmentation tasks (Fig. 1a). In MedSegDB, data samples are handled with an identical preprocessing protocol and organized by MedSegHierarchy, an anatomical hierarchy tailored for medical segmentation based on standard Radiology Lexicon (RedLex)[23] and radiologists' annotations. This hierarchy provides organization for the effective management and continuous expansion of the data (Fig. 1b).

Furthermore, we propose ConMoAE to introduce contextual knowledge in the model and provide the most semantic-related submodels for diverse tasks (Fig. 1c). Specifically, ConMoAE is composed of a Hierarchical Structure-Based Contextual Embedding Prior (HScEP) and a Context-Guided Mixture of Adapters Network (CMoAN). HScEP is a tree-structured embedding network that leverages the constructed hierarchy, MedSegHierarchy, as contextual knowledge that models semantic relationships among different tasks and groups semantically related tasks across diverse datasets. On the basis of the constructed hierarchy, HScEP initializes a contextual prior pool where each node within the tree is represented as an anatomy or a task embedding. CMoAN is developed on the basis of the Mixture of Experts (MoE)[24], composed of a set of adapter experts and a gate model, to adapt the MedSegX model to a specific context from various tasks. Taking contextual embedding as a condition, CMoAN dynamically activates a linear combination of different adapters to generate a semantic-related expert for a specific task, thus facilitating positive transfer.

Finally, we introduce a comprehensive evaluation scheme to analyse the performance of MedSegX in open-world medical segmentation, including ID settings, OOD settings and real-world validation. For the OOD setting, both zero-shot and data-efficient generalization are considered to simulate practical clinical deployment. We further conducted real-world validation on five clinically meaningful datasets in zero-shot and fine-tuning settings. Results demonstrate that MedSegX achieves state-of-the-art performance across various modalities and organ systems in the ID setting on a collection of 100 downstream tasks, as well as mitigating ambiguity in segmentation. Furthermore, MedSegX achieves outstanding performance in the OOD setting, outperforming generalist models in both zero-shot and data-efficient generalization capability. Further results demonstrate MedSegX's generalization capability over a vast range of medical segmentation tasks in real-world clinical scenarios.

Results

MedSegDB is a hierarchical medical segmentation database

The building of a large-scale ontology of medical images with annotated masks is a key resource for developing an advanced vision foundation model for medical segmentation, as well as for providing critical training and benchmarking data for such algorithms. While the amount of public medical images available for segmentation has increased substantially, they are mostly scattered, limited-size datasets and often focused on specific modalities, organs or lesions. Effective organization and utilization of such large-scale data have not been investigated.

In this study, we construct MedSegDB, an extensive, high-quality and organized database annotated from 129 public repositories and 5 in-house datasets (Supplementary Table 1). MedSegDB comprises 1,672,275 image–mask pairs from 804,305 images covering 10 medical imaging modalities, 39 major organs and tissues, and 111 segmentation tasks from various anatomical regions, including head and neck, torso, skeleton, blood vessel, and skin, representing the most comprehensive coverage of the human body for medical imaging segmentation (Fig. 1a, and Extended Data Figs. 1 and 2). Supplementary Tables 2–4 further summarize the statistics of MedSegDB.

To ensure the quality of the collected data, we adopt a unified preprocessing protocol to ensure their uniformity as well as address the partial labelling problem (Supplementary Fig. 1). Details of the preprocessing procedure are further summarized in Methods. To effectively organize the included data, we constructed a tree-structured anatomical hierarchy, MedSegHierarchy, built upon radiologists' annotations and RadLex[23]. RadLex is a publicly available terminology lexicon developed by the Radiological Society of North America, which provides a uniform standard for all radiology-related information. MedSegHierarchy is constructed using anatomical terms (entity nodes) and positional relations between anatomical regions based on 'Contains' and 'Has_Part' relations, providing a task-adaptive database with different semantics (Fig. 1b, see more details in Methods). By leveraging MedSegHierarchy, the preprocessed datasets are further systematically organized into the hierarchy by allocating them under the corresponding leaf nodes.

MedSegX design and evaluation settings

MedSegX is an enhanced medical segmentation generalist, which trains the foundation model like an expert radiologist by incorporating a Contextual Mixture of Adapter Experts (ConMoAE) to enhance its efficient adaptation in various segmentation tasks while mitigating negative transfer. MedSegX is developed on the basis of the architecture of SAM[15], a state-of-the-art (SOTA) vision foundation model in the general domain, by integrating ConMoAE with the Feed Forward Network (FFN) of each Transformer layer (Fig. 1c). This integration is lightweight, enabling quick fine-tuning.

ConMoAE is a novel paradigm that can introduce contextual knowledge from MedSegHierarchy into the MoEs and then provide separate parameters for conflicting tasks and generate semantically relevant submodels for diverse tasks to mitigate 'negative transfer'. Our proposed ConMoAE is model agnostic and can be pre-trained on various tasks jointly without performance degradation. Specifically, ConMoAE comprises two main components: HScEP and CMoAN.

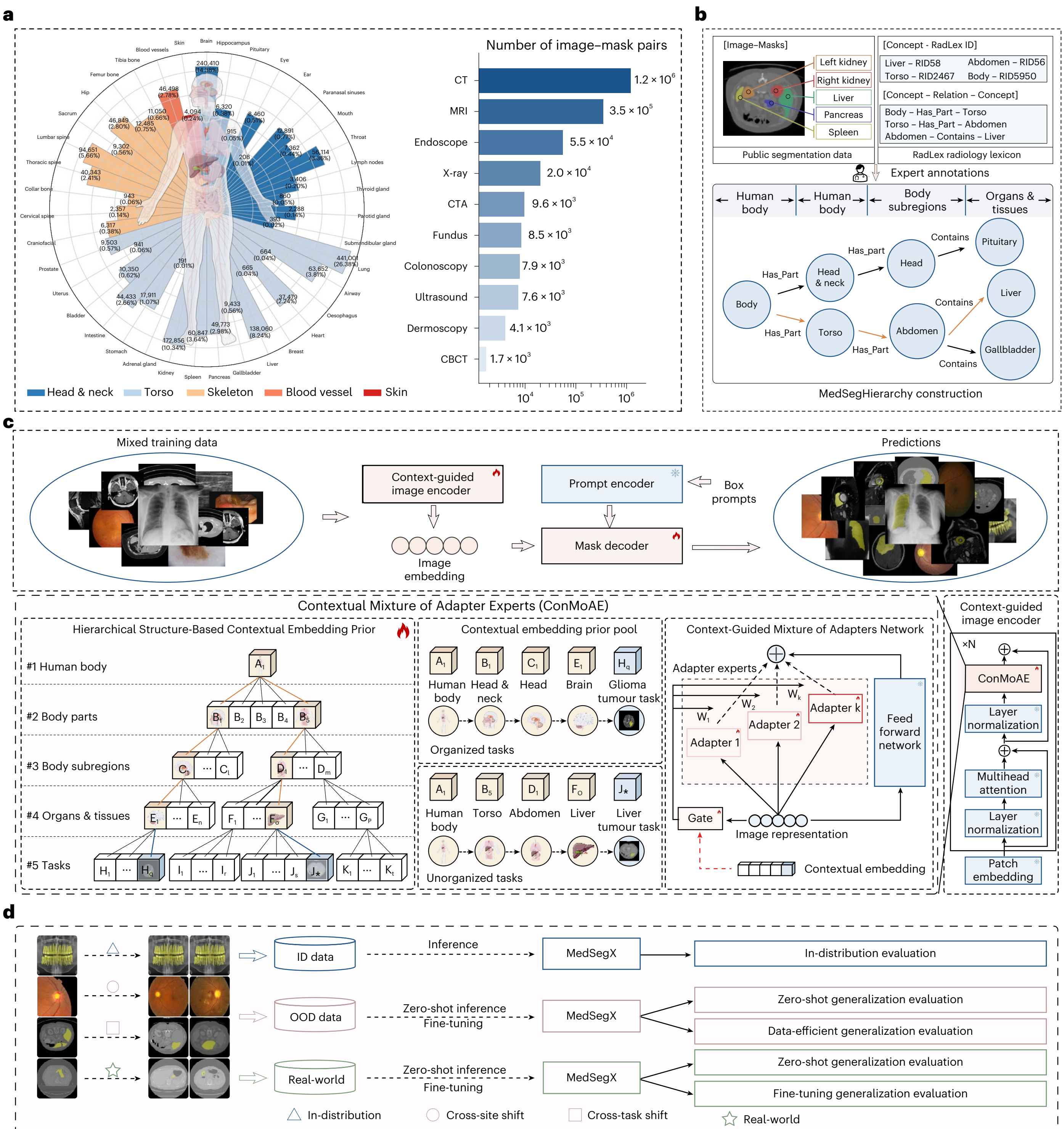

Fig. 1 | The tree-structured database MedSegDB and the development and evaluation of MedSegX. a, Overview of MedSegDB. MedSegDB covers 39 major organs and tissues, and 10 modalities. It is constructed and annotated primarily on the basis of public data, along with representative real-world datasets. CT, computed tomography; MRI, magnetic resonance imaging; CTA, CT angiography; CBCT, cone beam CT. **b**, Construction of MedSegHierarchy. MedSegDB is organized on the basis of MedSegHierarchy, which represents anatomy-related terminologies in RadLex as entity nodes and integrates them into the hierarchy on the basis of 'Has_part' and 'Contains' relations, thereby constructing a knowledge tree. **c**, Overall framework of MedSegX. MedSegX integrates ConMoAE with the Feed Forward Network of each Transformer layer in context-guided image encoder. ConMoAE is composed of HScEP (a Hierarchical Structure-Based Contextual Embedding Prior mapping explicit knowledge priors into contextual embedding) and CMoAN (a Context-Guided Mixture of Adapters Network consisting of a collection of adapter experts and a gating mechanism receiving contextual embeddings to provide task-adaptive models for diverse tasks). **d**, Comprehensive evaluation. MedSegX is evaluated on ID, OOD and real-world settings to establish the applicability of the model. For the OOD setting, both zero-shot and data-efficient performance evaluations are considered.

First, HScEP maps the task and anatomy nodes within MedSegHierarchy to contextual embedding, initializing a contextual prior pool. For a given image representation associated with a specific task, HScEP identifies a node path from the specific task node to the root anatomy node within the hierarchy. It then converts these nodes into embeddings to construct a contextual embedding prior pool. The contextual priors explicitly group semantically relevant tasks together, which could contribute to positive transfer, thereby mitigating the negative transfer impact caused by irrelevant tasks.

Subsequently, CMoAN receives contextual priors to dynamically activate a combination of different adapter experts. Specifically, the gate network takes the image representation with the contextual embedding as a condition to generate a gate vector, which determines the flexible activation of a linear combination of different adapters. This allows the production of separate submodels for diverse tasks, which mitigates the conflicts encountered across different tasks during pre-training, and activates the semantic-related submodels for specific tasks, eliminating negative transfer in downstream inference, thereby improving the performance of the generalist model across tasks at scale.

Although MedSegX depends on the explicit knowledge priors from MedSegHierarchy, it remains capable of being applied to tasks that have not been organized within the hierarchy. For unseen tasks, HScEP identifies the anatomical node closest to the target task and extends it to the root anatomy node. In addition, a new task node initialized by a fully zeroed vector is added for each unseen task in the last task layer, to generate a node path and corresponding contextual embedding. This process enables the model's generalization across diverse tasks, including those unseen during training (more details in Methods).

To comprehensively evaluate the open-world performance of MedSegX, we propose a systematic evaluation scheme with both ID and OOD settings (Fig. 1d and Supplementary Table 5, see more details in Methods). For the ID settings, the model is evaluated on ID test sets with 100 tasks, covering 10 modalities and 39 major organs and tissues. Specifically, we consider 2 OOD scenarios to simulate potential clinical deployment settings: (1) cross-site OOD evaluation: the model is evaluated on 18 anatomy-related tasks of 13 datasets collected from separate data sites, which reflects a variety of realistic distribution shifts owing to variations or discrepancies between the source and target datasets; and (2) cross-task OOD evaluation: the model is validated on 7 unseen categories of tumour segmentation tasks with different degrees of rarity, reflecting a bias due to the discrepancies between source and target segmentation objects. For the OOD settings, we included both zero-shot and data-efficient performance evaluation. Furthermore, to evaluate the clinical applicability of our MedSegX, we have constructed 5 real-world datasets of 4 hospitals in China, comprising 9 representative radiology tasks (see more details in Methods).

Performance of universal segmentation under ID settings

To evaluate MedSegX's performance, we conduct an ID evaluation on 100 tasks across 88 datasets, covering 327,732 image–mask pairs. The 100 tasks encompass anatomical structures and lesions across 5 major parts of the human body and 10 medical-imaging modalities (Fig. 2a and Supplementary Fig. 2). For comparison, we included 5 baselines: 3 traditional models, such as task-specific nnU-Net[9], U-Mamba[25] and universal DoDNet[13], all trained from scratch on the medical segmentation dataset; and 2 vanilla generalist models, SAM-Med2D[19] and MedSAM[18], both fine-tuned on medical images by vision foundation models.

In ID evaluation, we employ the Dice scores and 95% Hausdorff Distance (HD95) as evaluation metrics. Figure 2b and Supplementary Table 6 show the average performance of MedSegX and other baselines. We find that the performance of vision foundation models, such as MedSegX, SAM-Med2D and MedSAM, tends to surpass that of traditional models. Compared with other methods, MedSegX achieves the highest performance with a Dice score of 0.9109 (95% CI: 0.9105–0.9113) and HD95 of 4.372 (95% CI: 4.346–4.398) when averaging over all segmentation tasks. This significantly outperforms other vanilla generalist models, with MedSAM obtaining a Dice score of 0.8742 (95% CI: 0.8736–0.8747) and HD95 of 5.639 (95% CI: 5.609–5.669), and SAM-Med2D obtaining a Dice score of 0.8406 (95% CI: 0.8400–0.8413) and HD95 of 10.97 (95% CI: 10.90–11.03).

We further perform detailed analysis to investigate their performance when evaluated on various types of task. First, the segmentation tasks are categorized into anatomical structures and lesions (Fig. 2c). In segmentation tasks related to anatomical structures, MedSegX demonstrates superior performance with an average Dice score of 0.9336 (95% CI: 0.9332–0.9340), compared with the task-specific models (nnU-Net) as well as other vision foundation models such as MedSAM, with an improvement of 11.9% and 2.96%, respectively. For tasks related to more challenging lesion segmentations, MedSegX achieves an average Dice score of 0.7934 (95% CI: 0.7921–0.7947), surpassing the closest competitor by a substantial margin of 7.38%. Then, we evaluate the models' performance on segmentation tasks across 5 body parts, including the head and neck, torso, skeleton, blood vessel, and skin (Fig. 2d and Extended Data Fig. 3). The results demonstrate that MedSegX shows advantages in all tasks across 5 body parts. The detailed Dice scores and HD95 on each task of MedSegX and other baselines are further summarized in Supplementary Tables 7 and 8. We also evaluate the performance of models across 10 imaging modalities (Fig. 2e and Supplementary Tables 9 and 10). Notably, MedSegX surpasses other baselines across all modalities, including common modalities such as CT, MRI and X-ray, with Dice scores of 0.9397 (95% CI: 0.9393–0.9400), 0.8127 (95% CI: 0.8116–0.8139) and 0.8353 (95% CI: 0.8311–0.8395), respectively. We observe that MedSegX consistently outperforms all vanilla generalists by a large margin, demonstrating the potential of our Contextual MoEs for alleviating negative transfer.

The adaptability of MedSegX under varying conditions is also tested, including prompt types, model size scales and image resolutions within the ID setting. To extend MedSegX's applicability, we have introduced a point-based prompt model, which consistently achieves superior performance compared with other foundation models (Supplementary Table 11). Moreover, the evaluation of MedSegX based on SAM-Base (94 M), SAM-Large (312 M) and SAM-Huge (641 M) backbones demonstrates progressive improvements in Dice scores, achieving 0.9109 (95% CI: 0.9105–0.9113), 0.9169 (95% CI: 0.9165–0.9174) and 0.9180 (95% CI: 0.9176–0.9184), respectively (Supplementary Table 12). In addition, the evaluation conducted at lower (256 × 256) and higher (1,024 × 1,024) resolutions shows consistent Dice scores, demonstrating MedSegX's robustness across different image resolutions (Supplementary Table 13). To comprehensively assess our MedSegX model, we further conduct a comparison against 10 SOTA models across 10 public challenges from the ID set (Supplementary Fig. 3 and Supplementary Table 14). Evaluation metrics include Dice, HD95, Intersection over Union (IoU)[26], Mean Absolute Distance (MAD)[27], F2 score[26] and Recall[26], to provide a comprehensive comparison between MedSegX and SOTA models. The results show that MedSegX consistently outperforms other SOTAs across all metrics.

Performance on ambiguous tasks under ID settings

Model ambiguity refers to the uncertainty or lack of clarity that may arise in the segmentation tasks, due to insufficient or conflicting information within the input data or model architecture. For example, in SAM-based models[18,19] using prompting techniques[15], one input prompt may be ambiguous as it corresponds to several valid masks, leading to confusion in the model's predicted targets. Ambiguity directly influences how effectively the model addresses conflicts inherent in its predictions. Therefore, it is critical to evaluate a model's performance on ambiguous tasks and mitigate model ambiguity in segmentation.

We compare MedSegX with other SAM-based methods (SAM-Med2D and MedSAM) on 3 ambiguous tasks: (1) brain whole

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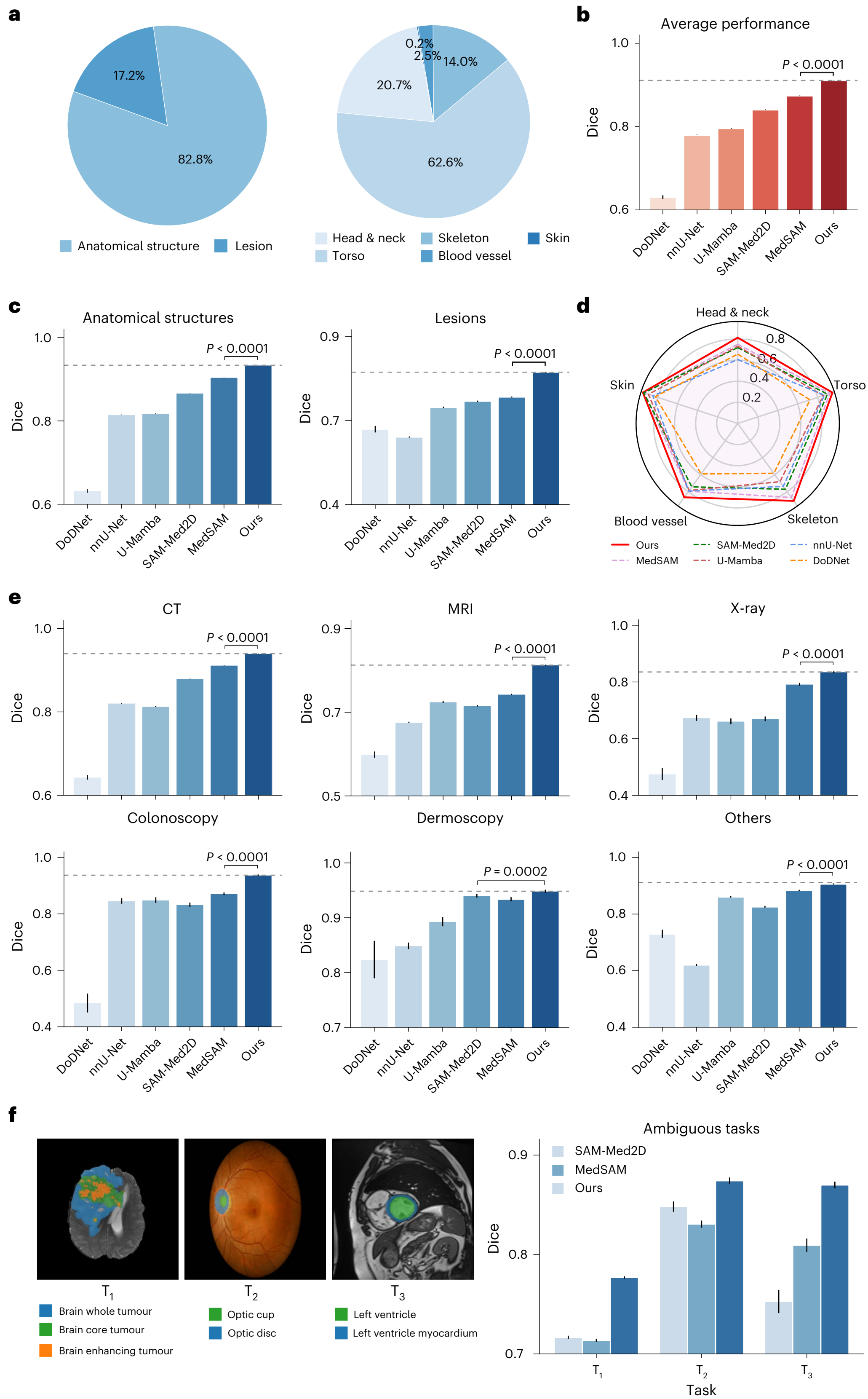

Fig. 2 | Performance on ID evaluation. a, Statistics of the ID test set. The proportion of ID test data on task types and five body parts. **b**, Average performance of 6 models on all tasks (n = 327,732). **c**, Average performance of the 6 models on anatomical (n = 271,493) and lesion-related (n = 56,239) tasks. **d**, Performance comparison of the 6 models on tasks related to 5 body parts: head and neck, torso, skeleton, blood vessel and skin. **e**, Average performance of the 6 models on CT (n = 238,686), MRI (n = 67,357), X-ray (n = 3,758), colonoscopy (n = 1,588), dermoscopy (n = 818) and other modalities (n = 15,525). **f**, Performance comparison of SAM-Med2D, MedSAM and MedSegX on 3 ambiguous tasks (n = 42,615). T_1: brain whole tumour, brain core tumour and brain enhancing tumour; T_2: optic cup and optic disc; T_3: left ventricle and left ventricular myocardium. P values were calculated using two-sided t-test. Bar graphs indicate the mean ± 95% CI. The dashed horizontal lines in **b**, **c** and **e** represent the Dice scores obtained by MedSegX.

tumour, core tumour and enhancing tumour segmentation; (2) optic disc and cup segmentation; and (3) left ventricle and ventricular myocardium segmentation. In these 3 tasks, ambiguous task overlapping exists, which leads to multiple targets corresponding to the same input prompt. As shown in Fig. 2f and Supplementary Table 15, MedSegX demonstrates robustness to ambiguous tasks with enhanced ambiguity awareness of the Dice score of 0.7961 (95% CI: 0.7946–0.7976), 0.9177 (95% CI: 0.9134–0.9219) and 0.9124 (95% CI: 0.9082–0.9166) on the 3 ambiguous tasks compared with other SAM-based baselines, with MedSAM obtaining 0.7171 (95% CI: 0.7154–0.7188), 0.8632 (95% CI: 0.8586–0.8677) and 0.8366 (95% CI: 0.8282–0.8451), respectively, and SAM-Med2D obtaining 0.7208 (95% CI: 0.7188–0.7229), 0.8852 (95% CI: 0.8787–0.8917) and 0.7659 (95% CI: 0.7513–0.7804), respectively. The results further highlight the effectiveness of our contextual knowledge-enhanced learning in addressing task ambiguity.

Generalization capability under cross-site OOD settings

Cross-site shifts commonly occur in varying data collection processes such as diverse data-acquisition devices or clinical demographics[28], which are quite common in real-world clinical scenarios and can be hard to spot[29]. Therefore, models may not generalize well in OOD settings where the training data are, in some subtle manner, not representative of the target task data distribution.

Here we extend our evaluation of the generalizability of our MedSegX and other 4 baselines to cross-site settings. This cross-site OOD evaluation included 18 anatomy-related segmentation tasks from 13 datasets, which were collected from separate data sites, distinct from the ones used for training (Fig. 3a). Taking the lung segmentation task on X-ray as an example, we incorporated the JSRT dataset[30] during the training phase, which includes cases from healthcare facilities in Japan and the United States. In contrast, cases for the same task included in the OOD set were collected from the SZCXR dataset[31] in China, leading to a variety of distribution shifts in population, behaviour and data collection devices.

For the OOD settings, zero-shot and data-efficient performance evaluation are both considered. The performance of MedSegX and other methods under cross-site shift with the zero-shot setting is shown in Fig. 3b and Supplementary Tables 16–18. In this setting, a model is directly applied to OOD test sets without requiring further labelled examples for supervised learning or fine-tuning, to assess the model's generalization capability to unseen tasks without any task-specific training. The results demonstrate that MedSegX achieves the best zero-shot performance, with a Dice score of 0.8733 (95% CI: 0.8706–0.8760) and HD95 of 4.736 (95% CI: 4.557–4.915). This significantly outperforms another state-of-the-art model MedSAM, with a Dice score of 0.7296 (95% CI: 0.7233–0.7359) and an HD95 of 10.14 (95% CI: 9.736–10.55). Taking gallbladder segmentation on CT as an example, we find that MedSegX achieves the best performance of 0.8544 (95% CI: 0.8483–0.8605), surpassing the closest competitor by a substantial margin of 25.49%. These results indicate MedSegX's potential generalizability to other data-acquisition devices or populations with efficiency.

In addition, we investigate the data-efficient generalizability of the models, by conducting fine-tuning on varying percentages (5%, 15%, 25%, 50% and 100%) of the OOD fine-tuning set and then evaluating on OOD test sets. This aims to evaluate the model's capacity to effectively generalize to new clinical scenarios where labelled data are scarce. Specifically, Fig. 3c illustrates the result using only 5% OOD of fine-tuning data which presents a more challenging scenario. Notably, even with the limited 5% of the fine-tuning data, MedSegX exhibits a significantly superior performance compared with other methods, achieving an average Dice score of 0.9070 (95% CI: 0.9005–0.9135). This score surpasses that of the next best-performing model by a margin of 5.92% (Fig. 3c). Similar results were also observed under other different proportions of OOD fine-tuning data (15%, 25%, 50% and 100%; Extended Data Fig. 4).

Figure 3d, Extended Data Fig. 5 and Supplementary Tables 19–23 provide additional insight into the models' data efficiency with a continuous proportion of fine-tuning data across 18 tasks. The results highlight that with as little as 5% to 25% of the fine-tuning data, MedSegX can achieve consistently comparable or superior performance compared with other generalist segmentation methods fine-tuned with 100% OOD data, such as MedSAM (Fig. 3d) and SAM-Med2D (Extended Data Fig. 5), thus demonstrating MedSegX's remarkable ability for data-efficient generalization to other data-acquisition devices or populations.

Generalization capability under cross-task OOD settings

In addition to cross-site evaluation that focuses mainly on data shifts due to varying data collection processes, we further evaluate the performance of MedSegX in a cross-task setting. Cross-task generalization requires the model to perform segmentation on new targets that were unseen during training. Here we investigate the models to segment 7 previously unseen categories of body tumours with different degrees of rarity[32]. This includes common tumours in the colon, kidney, liver and lung, along with relatively common tumours in the pancreas and prostate, and rare tumours such as vestibular schwannoma in the vestibulocochlear nerve (Fig. 4a). The evaluation is performed on 10 datasets for both zero-shot and data-efficient generalization settings.

Figure 4b and Supplementary Tables 24–26 show the zero-shot performance of our MedSegX and other baselines under cross-task shifts. Results demonstrate that MedSegX achieves a higher average Dice score of 0.7691 (95% CI: 0.7624–0.7758), compared with that of the second best-performing model SAM-Med2D with 0.6886 (95% CI: 0.6807–0.6966). We also observe a decline in performance on tumour-related tasks across all models, as tumour segmentation is typically more challenging than anatomical segmentation due to its heterogeneous characteristics in shape, texture and intensity. However, MedSegX still remains significantly superior to the second-best model, with an average Dice score margin of 8.05%.

Furthermore, we conduct data-efficient generalization evaluation on our MedSegX and other baselines. When employing various fractions of OOD data (5%, 15%, 25%, 50% and 100%) for fine-tuning, we find that MedSegX exhibits better generalization capabilities compared with other models (Fig. 4c and Extended Data Fig. 6). It is worth noting that MedSegX enhances the average performance from 0.7691 (95% CI: 0.7624–0.7758) to 0.8032 (95% CI: 0.7966–0.8098) with only a small number of fine-tuning data (5%) (Fig. 4c). Similar results were also achieved under other different proportions of OOD fine-tuning data (15%, 25%, 50% and 100%; Extended Data Fig. 6). Figure 4d, Extended Data Fig. 7 and Supplementary Table 27–31 further provide details on data efficiency of MedSegX and other baselines. The results demonstrate that MedSegX requires only 5–25% of fine-tuning data to match or surpass the performance of SOTA baselines fine-tuned with 100% OOD data, demonstrating data-efficient generalization and adaptability to unseen tasks.

Clinical applicability on real-world radiology tasks

To further evaluate the clinical applicability of MedSegX on real-world data, we investigate the segmentation performance of MedSegX on five representative datasets. Specifically, we conduct colon tumour segmentation on the colon cancer dataset, stomach tumour segmentation on the stomach cancer dataset, lung tumour segmentation on the lung cancer dataset, hemangioma and hepatocellular carcinoma segmentation on the liver masses dataset, as well as consolidation, ground-glass opacity, interstitial thickening and pleural effusion segmentation on the pneumonia dataset.

We first evaluate the zero-shot performance of MedSegX and other baselines on real-world datasets (Fig. 5a, Extended Data Fig. 8a

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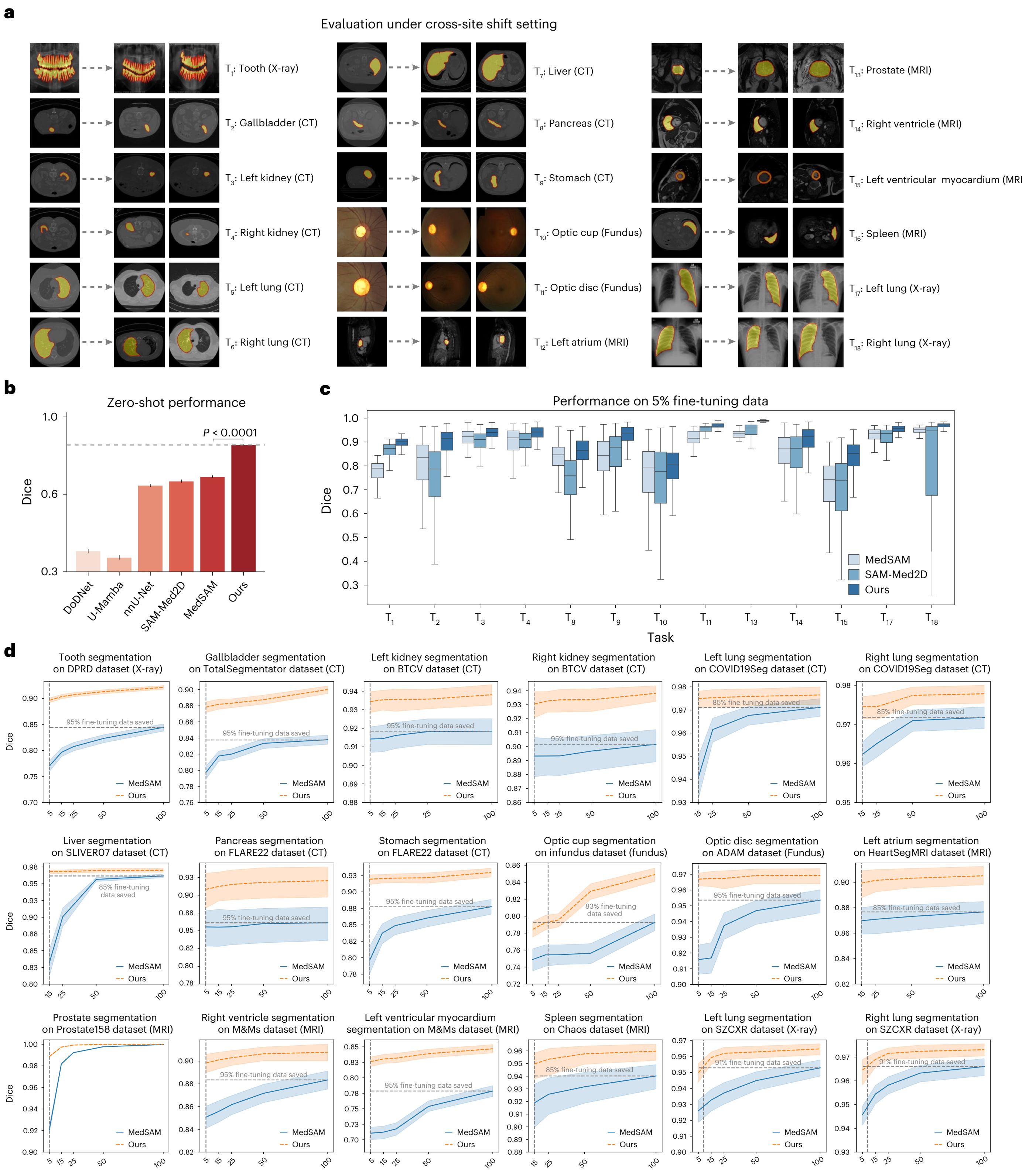

Fig. 3 | Performance under cross-site shift setting. a, Details of tasks. We perform cross-site evaluation on 18 anatomy-related tasks, including tooth (X-ray), gallbladder (CT), left kidney (CT), right kidney (CT), left lung (CT), right lung (CT), liver (CT), pancreas (CT), stomach (CT), optic cup (Fundus), optic disc (Fundus), left atrium (MRI), prostate (MRI), right ventricle (MRI), left ventricle (MRI), spleen (MRI), left lung (X-ray) and right lung (X-ray). **b**, Zero-shot performance evaluation of MedSegX and other models on 18 cross-site tasks ($n = 5{,}801$). *P* value was calculated using two-sided *t*-test. The bar graph indicates the mean ± 95% CI. The dashed horizontal line represents the Dice score obtained by MedSegX. **c**, Generalization evaluation of MedSegX and other models on 13 cross-site tasks with only 5% fine-tuning data ($n = 5{,}801$). T_5, T_6, T_7, T_{12} and T_{16} lack the 5% fine-tuning cases and are thus not presented. Each box shows the quartiles of the distribution, with the centre as the median, the minimum as the first quartile, and the maximum as the third quartile; the whiskers extend to the farthest data point that lies within 2× the interquartile range (IQR) from the nearest quartile. **d**, Data-efficient performance evaluation on 18 cross-site tasks. 95% CIs are shown by the shaded area.

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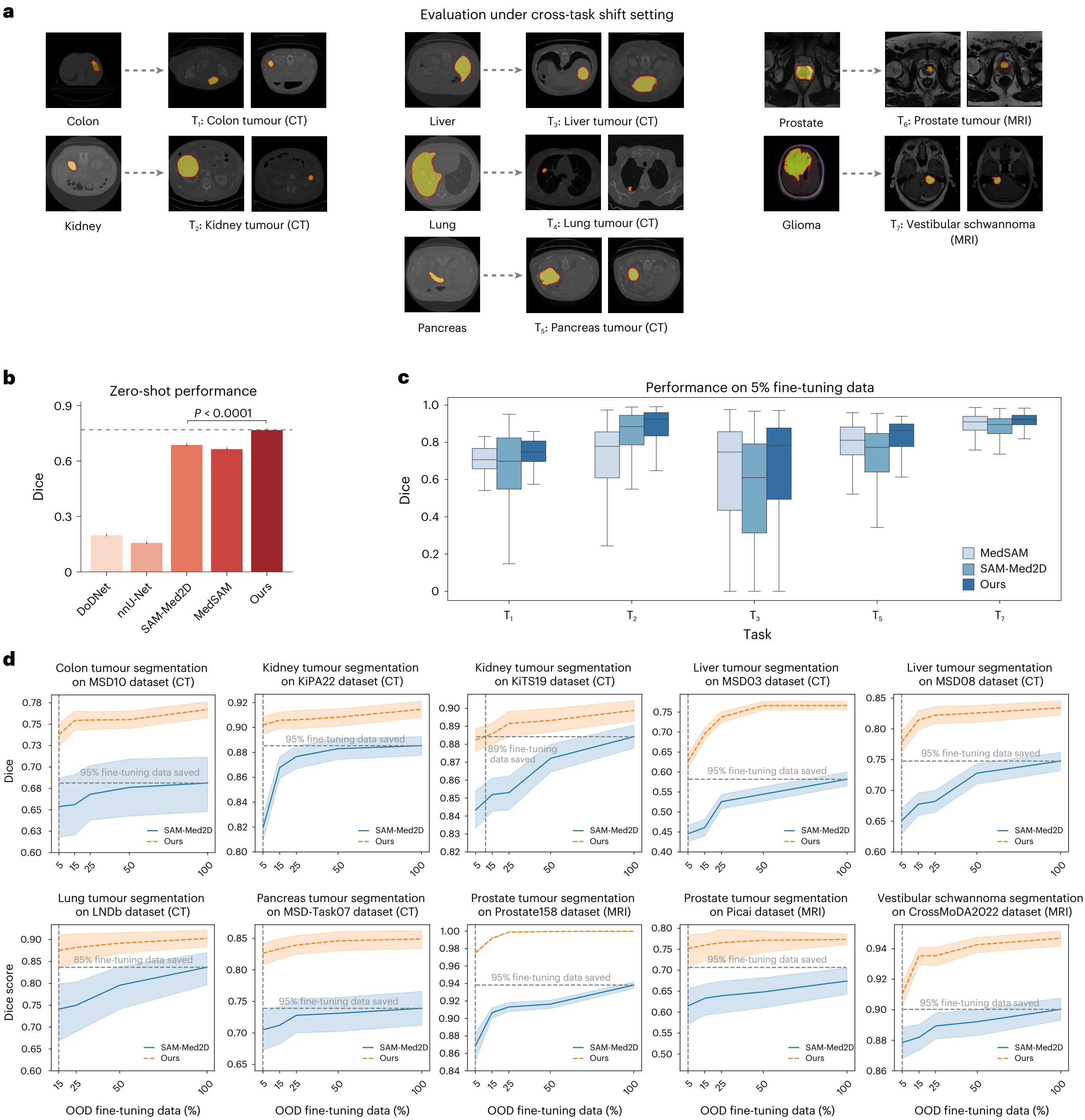

Fig. 4 | Performance under cross-task shift setting. a, Details of tasks. We perform cross-task evaluation on 7 unseen categories of body tumours, including colon tumour (CT), kidney tumour (CT), liver tumour (CT), lung tumour (CT), pancreas tumour (CT), prostate tumour (MRI) and vestibular schwannoma (MRI). **b**, Zero-shot performance evaluation of 5 models on 7 unseen categories of body tumours ($n = 4{,}095$). P value was calculated using two-sided t-test. The bar graph indicates the mean ± 95% CI. The dashed horizontal line indicates the Dice score of MedSegX. **c**, Generalization evaluation of our MedSegX, MedSAM and SAM-Med2D on 5 tasks (T_1, T_2, T_3, T_5, T_7) under cross-task shift setting with only 5% fine-tuning data ($n = 4{,}095$). No 5% fine-tuning cases were observed in T_4 and T_6; hence they are not included in the presentation. Each box shows the quartiles of the distribution, with the centre as the median, the minimum as the first quartile, and the maximum as the third quartile; the whiskers extend to the farthest data point that lies within 2× the IQR from the nearest quartile. **d**, Data-efficient performance evaluation on 7 tasks under cross-task shift setting. 95% CIs are shown by the shaded area.

and Supplementary Tables 32–34). The average results demonstrate MedSegX's superior generalization capability, outperforming other SOTA baselines (Extended Data Fig. 8a). While all models exhibit a decline in performance in zero-shot clinical scenarios, MedSegX consistently surpasses other models across different datasets (Fig. 5a and Supplementary Tables 32 and 33). Specifically, it achieves substantial Dice margins of 9.91%, 7.79%, 27.5%, 13.4% and 16.7% compared with the next best-performing models in the colon cancer dataset, stomach cancer dataset, lung cancer dataset, liver masses dataset and pneumonia dataset, respectively (Supplementary Table 34).

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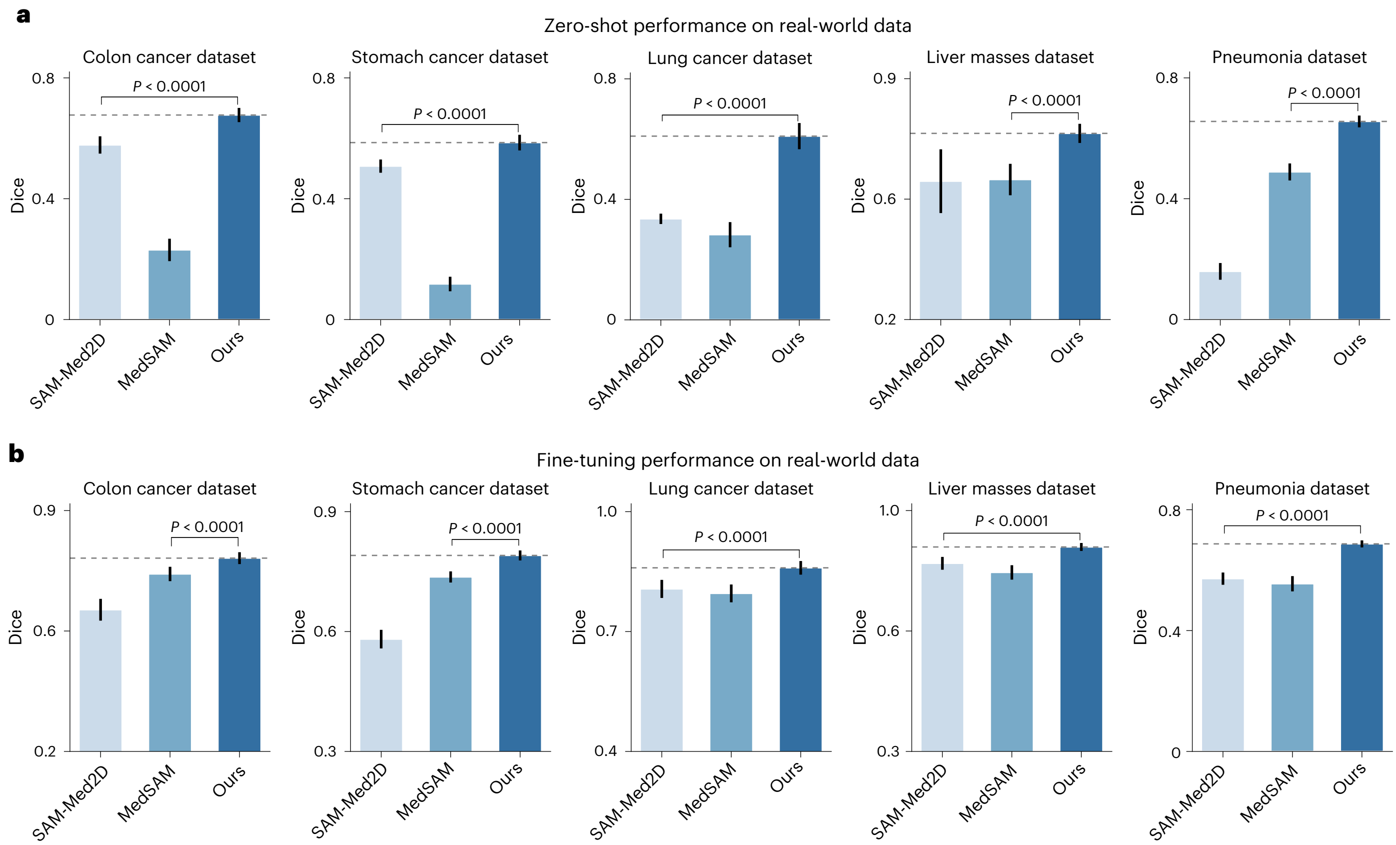

Fig. 5 | Performance evaluation under real-world settings. a, Zero-shot performance evaluation across 5 real-world datasets, including colon cancer ($n = 284$), stomach cancer ($n = 268$), lung cancer ($n = 146$), liver masses ($n = 77$) and pneumonia ($n = 175$). **b**, Fine-tuning performance evaluation on 5 real-world datasets, including colon cancer ($n = 284$), stomach cancer ($n = 268$), lung cancer ($n = 146$), liver masses ($n = 77$) and pneumonia ($n = 175$). P values were calculated using two-sided t-test. Bar graphs indicate the mean ± 95% CI. The dashed horizontal lines in **a** and **b** represent the Dice scores obtained by MedSegX.

We further evaluate the models' fine-tuning performance (Fig. 5b, Extended Data Fig. 8b and Supplementary Tables 35–37). MedSegX demonstrates enhanced clinical applicability across various scenarios, such as lung cancer dataset, stomach cancer dataset, colon cancer dataset and liver masses dataset, with the best Dice scores of 0.8595 (95% CI: 0.8424–0.8765), 0.7905 (95% CI: 0.7780–0.8030), 0.7618 (95% CI: 0.7445–0.7791) and 0.8942 (95% CI: 0.8827–0.9058), respectively (Fig. 5b and Supplementary Table 37). Notably, for more challenging pneumonia-related tasks characterized by diffuse distribution, while the models show a decline in performance, MedSegX still outperforms the second-best model by a significant margin, with an average Dice advantage of 11.5% (Supplementary Table 37). These results further highlight MedSegX's applicability in complex clinical scenarios.

Visualization analysis for MedSegX

Furthermore, we conduct a visualization analysis on 8 randomly selected OOD examples (Extended Data Fig. 9): 2 examples from cross-site tasks (T_1–T_2) and 6 examples from unseen categories of body tumours (T_3–T_8). Extended Data Fig. 9 illustrates the segmentation results obtained by MedSegX, MedSAM and SAM-Med2D fine-tuned with 100% OOD data. While all models perform well on simple targets with regular shapes, such as left ventricular myocardium segmentation (T_1), MedSegX achieves the best performance with more reliable predictions. Moreover, for complex anatomical regions with irregular shapes, such as teeth segmentation (T_2), MedSegX generates more accurate target boundaries. Regarding tumour segmentation (T_3–T_8), which is inherently challenging due to variability in size, shape, texture and appearance, MedSegX exhibits fewer false positive and false negative predictions across tumours in the colon (T_3), kidney (T_4), liver (T_5), lung (T_6), pancreas (T_7) and prostate (T_8). In summary, the results demonstrate MedSegX's capability in handling both simple and complex segmentation tasks across various anatomical structures and tumour types, underscoring its robustness and versatility in medical image analysis.

Ablation study

To analyse the effectiveness of our proposed approach, we conduct a detailed ablation experiment on different components: (1) Adapter (single): only one adapter is integrated with the FFN of each Transformer layer in the image encoder; (2) Adapter with CMoAN: the CMoAN, without HScEP, is integrated into the image encoder; (3) Adapter with CMoAN and HScEP: CMoAN, combined with HScEP, is integrated in the image encoder, which introduces contextual knowledge into the model. The corresponding ablation results are summarized in Supplementary Table 38. Both ID and OOD settings are considered. The results show that without the guidance of contextual knowledge, Adapter with CMoAN improves the model performance of Adapter (single) slightly. When contextual knowledge from HScEP is incorporated into the model (Adapter with CMoAN and HScEP), there is a significant performance improvement in both ID and OOD scenarios ($P < 0.05$), from 0.9060 (95% CI: 0.9056–0.9065) to 0.9109 (95% CI: 0.9105–0.9113) under ID settings, from 0.8270 (95% CI: 0.8223–0.8316) to 0.8733 (95% CI: 0.8706–0.8760) under cross-site settings, and from 0.7545 (95% CI: 0.7476–0.7613) to 0.7691 (95% CI: 0.7624–0.7758) under cross-task settings. In addition, we have performed a visualization analysis of different components under both ID and OOD settings (Extended Data Fig. 10), which further highlights the effectiveness of the combination of HScEP and CMoAN in enhancing model performance.

Applicability of three-dimensional (3D) segmentation

To further validate the effectiveness of MedSegX, we extend MedSegX to MedSegX3D for 3D segmentation by incorporating volumetric information from 3D medical data (see more details in Methods). For comparison, we include 7 baselines: 4 task-specific models, such as 3D nnU-Net[9], 3D U-Mamba[25], nnFormer[33] and MISSU[34], all trained from scratch on the medical segmentation dataset; 3 vanilla generalist models, SAM-Med3D[19] fine-tuned on medical images based on SAM[15] and Medical SAM 2 (ref. 35), MedSAM2 (ref. 36) based on improved SAM, called SAM 2 (ref. 37). We conduct experiments on 3D data from the ID set. Figure 6a and Supplementary Table 39 show the average performance of MedSegX3D and the other 3D baselines. Compared with the other methods, MedSegX3D achieves the highest performance with an average Dice score of 0.8523 (95% CI: 0.8485–0.8562) and HD95 of 3.072 (95% CI: 2.988–3.157). This significantly outperforms the best task-specific model, 3D nnU-Net, with a Dice score of 0.7983 (95% CI: 0.7925–0.8041) and HD95 of 6.357 (95% CI: 5.982–6.732), and the vanilla generalist model, SAM-Med3D, with a Dice score of 0.5270 (95% CI: 0.5210–0.5330) and HD95 of 12.23 (95% CI: 12.01–12.45). Then, we evaluate the models' performance on 3D segmentation tasks across different body parts and observe that MedSegX3D consistently outperforms all task-specific small models and vanilla generalists (Fig. 6b). Furthermore, we investigate detailed segmentation performance on commonly used 3D modalities, including CT, MRI, CTA and CBCT (Fig. 6c and Supplementary Tables 40 and 41). Notably, MedSegX3D surpasses the second-best baselines across all modalities, achieving substantial Dice improvements of 4.62%, 5.73%, 7.46% and 5.88% in CT, MRI, CTA and CBCT, respectively. These results further demonstrate the extended effectiveness of MedSegX on 3D segmentation, indicating its potential applicability in 3D clinical scenarios.

Discussion

Despite the progress in vision foundation models[15] for general medical image segmentation[18,19], there are still challenges that have not yet been overcome. To address these issues, our work has several contributions. First, we have constructed MedSegDB, a large-scale medical segmentation database with high-quality and standardized labelling annotations, organized using MedSegHierarchy, a tree-structured hierarchy to enhance data scalability. Second, to mitigate negative transfer issues in medical vision foundation models, we propose a ConMoAE strategy, leveraging explicit knowledge priors from MedSegHierarchy to group semantically relevant tasks, and providing separate and semantic-related submodels for diverse tasks; thus, we mitigate task conflicts and facilitate positive transfer during simultaneous training of large-scale data. In addition, we conduct a comprehensive evaluation including ID setting with 100 tasks, OOD setting involving site shifts and task shifts across 25 tasks, and real-world setting on 5 representative datasets. Experiments demonstrate that our generalist model can be pre-trained on multiple tasks jointly without performance degradation while maintaining its generalizability to novel tasks.

Recently, vision foundation models have advanced the field of segmentation, providing a more versatile task-agnostic solution compared with previous 'specialist' models for a wide range of applications. Nevertheless, the direct application or pre-training of these generalist models on various datasets presents challenges. Some research observed that discrepancies between source data and downstream target data during pre-training often leads to the negative transfer phenomenon[20,21]. Unified modelling to process different tasks using shared parameters would face 'task conflicts', where different tasks with shared parameters may conflict with each other[22]. These challenges highlight the need for an enhanced generalist technique to strike a balance between generalization and specialization for open-world medical segmentation. Observing the training process of radiologist experts indicates that their expertise arises from previous knowledge and understanding of anatomical structures across a diverse range of medical images spanning body regions, diseases and modalities. Inspired by this, we propose a task-adaptive generalist, MedSegX, which includes a novel Contextual Mixture of Adapter Experts (ConMoAE) strategy and a multi-expert architecture to provide task-adaptive submodels for different input tokens based on MoE[24].

Current foundation models are generally trained on vast amounts of data to achieve their high performance and generalization capabilities. While several studies have investigated the collection of large-scale data for universal medical segmentation[18,38,39], they lack a unified label system to maintain consistency across diverse annotations, which compromises the quality of the labels. For example, due to distinct clinical objectives, different datasets have different and even incomplete object annotations: for abdominal segmentation, AbdomenCT-1K[40] contains annotations for 4 organs, while AMOS22 (ref. 41) annotates 15 targets. We also observe that class definitions can vary depending on the clinical targets: in the AbdomenCT-1K[40] dataset, the liver is annotated as a whole organ, whereas in the LiTS[42] dataset, the same region is divided into two categories (liver and liver tumour). Moreover, the same anatomical targets from different datasets are named inconsistently: the *i*th lumbar vertebrae in both TotalSegmentator[43] and MRSpineSeg[44] are named with the format 'lumbar vertebrae *i*' and 'L*i*', respectively. Therefore, we employ a unified preprocessing protocol to standardize annotations and consolidate identical tasks, resulting in 111 high-quality labels for utilization by both researchers and developers.

Furthermore, existing research[18,38,39] lacks consideration of effective ways of organization, thereby hindering the future scalability of the data. Inspired by previous work, such as ImageNet[45], VisualGenome[46] and Chest ImaGenome[47], we collect data from 129 public repositories and 5 in-house datasets to construct MedSegDB, a medical segmentation database of 1,672,275 image–mask pairs, which is organized into MedSegHierarchy, a tree-structured hierarchy built upon RadLex[23] and radiologists' annotations. On the basis of the hierarchy, semantically related tasks are aggregated together to provide knowledge contextual priors, which could further enhance our proposed foundation model.

Despite the advances presented in our study, our study still has aspects that need further investigation in the future. First, while we have focused on medical imaging analysis by assisting radiologists within the workflow, we plan to further incorporate pathological knowledge (for example, OncoTree) to extend our MedSegX model. Specifically, we could integrate pathology modality embedding nodes into the existing medical modality nodes. To better leverage OncoTree's pathological knowledge, we could align its hierarchical structure—tissues and tumours—with our proposed MedSegHierarchy and update pathological data in the corresponding nodes. This enhancement will provide MedSegX with complementary insights at the cellular level, thus enhancing the model's applicability. Second, class imbalance is a common issue encountered in clinical scenarios, such as targets with varying sizes and shapes. Although we have addressed this by incorporating class weighting in the cross-entropy loss and by collecting a substantial dataset of small targets to enhance feature representation, this can be further improved. Future strategies may include techniques such as data oversampling or undersampling. Furthermore, quantifying the effect of negative transfer (for example, negative transfer gap, NTG) represents an interesting direction. We plan to explore the potential for early detection of NTG before testing. Strategies such as intertask affinity[48] could be used to analyse how the gradient of one task affects the loss of another, providing a basis for quantifying the impact of negative transfer between tasks during training. In addition, we will incorporate more evaluation metrics (for example, Dice, HD, F2 score) for a more comprehensive NTG. In future work, additional real-world datasets covering more data modalities or tasks will be included to fully evaluate the clinical use of our proposed method.

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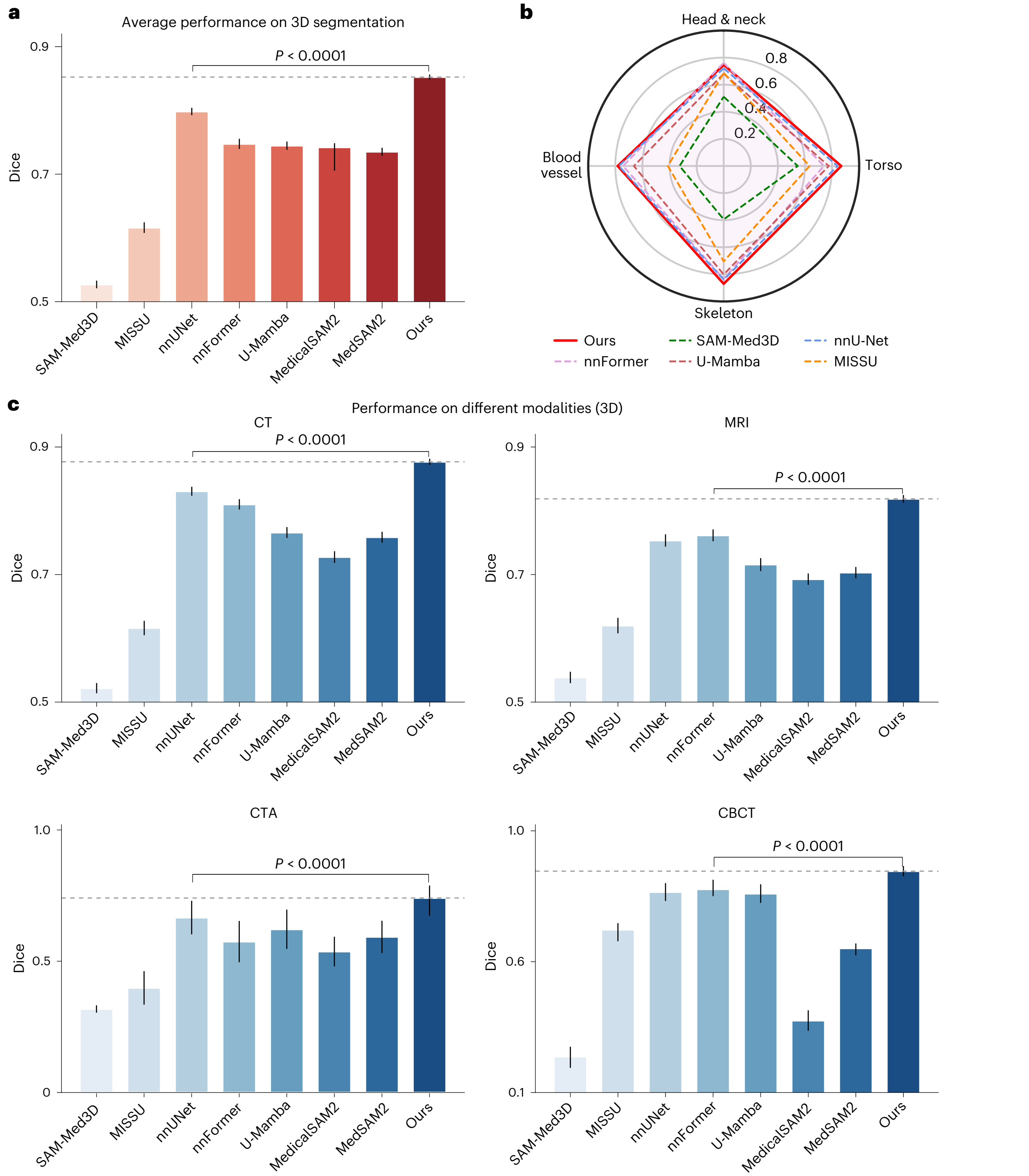

Fig. 6 | Performance evaluation on 3D segmentation. **a**, Average performance of 8 models on 3D segmentation tasks ($n = 24{,}217$). **b**, Performance comparison of the 8 models on tasks related to 4 body parts: head and neck, torso, skeleton and blood vessel. **c**, Average performance of the 8 models on 3D segmentation across CT ($n = 14{,}271$), MRI ($n = 9{,}734$), CTA ($n = 182$) and CBCT ($n = 30$) modalities. P values were calculated using two-sided t-test. Bar graphs indicate the mean ± 95% CI. The dashed horizontal lines in **a** and **c** represent the Dice scores obtained by MedSegX.

In summary, we have developed MedSegX on the basis of a vision foundation model, which is trained on an extensive, high-quality and organized database, as well as leveraging ConMoAE for generalist models while mitigating task conflicts and negative transfer. We constructed MedSegHierarchy, a tree-structured hierarchy for medical imaging that is built upon publicly available terminology lexicon and radiologists' annotation. We further propose ConMoAE, a novel adapter-based MoE architecture designed to activate the task-adaptive model parameters based on the hierarchy as context. We conduct stepwise evaluations in both ID and OOD settings, assessing the model's generalization capability in potential real-world deployment. The results demonstrate that MedSegX consistently achieves optimal performance across various modalities and organ systems while also maintaining sufficient effectiveness in OOD environments. This underscores its potential to expedite the development and deployment of a general medical segmentation model.

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Methods

Datasets collection and data preprocessing

Large-scale and high-quality data play a pivotal role in the development of foundation models. In our study, we conducted an extensive collection of 129 medical segmentation datasets sourced from SA-Med2D-20M[39] and various medical repositories, including the Cancer Imaging Archive (TCIA)[49], Grand-Challenge, Kaggle, Synapse, CodaLab, GitHub and segmentation challenges hosted by the Medical Image Computing and Computer Assisted Intervention Society (MICCAI). It is worth noting that some public datasets, such as KiTS19 (ref. 50) and KiTS21 (ref. 51), are not mutually exclusive. To ensure no overlap between datasets, we thoroughly examined the sources of each dataset, including relevant literature, challenge websites and any available official documentation to identify potential overlaps. After a detailed review, we manually excluded any datasets identified as containing potential duplicate cases.

Moreover, in clinical practice, medical imaging protocols vary among different modalities, thereby leading to the highly variable intensities of medical images. There exists heterogeneity among different datasets, such as inconsistent labelling standards and varying data resolutions, with a large effect on the development of AI-based models[38]. To ensure uniformity and compatibility of collected medical data, we preprocessed all image–mask pairs with a unified preprocessing protocol, which consists of four major steps, including intensity normalization, label standardization, unqualified data exclusion and data resizing. First, to ensure consistent image intensities, we employed self-min-max normalization using $\lceil \frac{x-x_{\min}}{x_{\max}-x_{\min}} \times 255 \rceil$ to transform raw image pixel intensities into the range of $[0, 255]$, where $x_{\min}$ and $x_{\max}$ represent the minimum and maximum pixel values in the raw image, respectively. Subsequently, we employed a standardized labelling protocol to enhance annotation uniformity: (1) multilabel annotations were converted into binary-label annotations by separating foreground targets, ensuring that each mask represents only one target class; (2) annotations were revised to resolve conflicts between overlapping classes. Specifically, when lesions occurred within an organ, the annotation encompassed both the organ and the lesion; and (3) each target was assigned a unified medical terminology to ensure consistent class definitions for identical objects. Furthermore, rigorous inclusion criteria were implemented to maintain data quality. Specifically, we excluded data samples with incomplete images, missing labels or tiny annotated targets (where the sum of target pixel values is less than 100). Finally, all images and corresponding masks were resized to a uniform size of 256×256 to align with model requirements. For 3D volumetric images, each slice was resized to 256×256along the primary viewing plane, with the channel of image slices repeated three times. For 2D data, images were resized to $256 \times 256 \times 3$, while corresponding masks were resized to 256×256.

Construction of MedSegHierarchy and MedSegDB

Although current foundation models are trained on the basis of large open datasets, the low-quality and semantically irrelevant datasets can substantially impact performance, leading to issues such as overfitting, bias and unfairness, ultimately resulting in poor generalization in real-world applications.

In this study, we constructed MedSegHierarchy, a tree-structured hierarchy built on RadLex[23], a publicly available terminology lexicon, developed by the Radiological Society of North America (RSNA), that provides a uniform standard for all radiology-related information. We concentrated on 56 anatomy-related terminologies in RadLex as entity nodes and integrated them into the hierarchy on the basis of 'Has_part' and 'Contains' relations with the assistance of imaging experts' annotations, thereby constructing a knowledge tree including the human body, body parts, body subregions, and organs and tissues (Fig. 1b). In addition, we mapped all terminologies in the hierarchy to a RadLex Identifier (RID) to widen future application of our constructed hierarchy. By fully exploiting the tree-structured hierarchy, we systematically organized the included data by allocating them into corresponding leaf nodes in the constructed hierarchy, thus constructing a medical segmentation database, MedSegDB. Within each location, data samples for the same tasks were grouped together, which could bring positive transfer to each other.

ID and OOD sets

To develop and validate the models, public data in MedSegDB were divided into 3 sets (Extended Data Fig. 2a and Supplementary Table 5), including an ID set incorporating 1,614,256 image–mask pairs of 100 segmentation tasks, and 2 OOD sets composed of a cross-site set with 32,282 image–mask pairs of 18 tasks and a cross-task set with 20,155 image–mask pairs of 7 tasks. Specifically, all cases in the ID set were split randomly into mutually exclusive sets for pre-training and evaluation at a ratio of 8:2. In the OOD set, all cases were split randomly into a fine-tuning set and a test set at a ratio of 8:2. Within the fine-tuning set, we divided the cases into different proportions (5%, 15%, 25%, 50%, 100%), further validating the data-efficient generalization capability of models. To achieve a mutually exclusive split on a per-patient basis, we assigned a unique identifier to each patient and grouped their images separately. On the basis of these groups, images from a single patient were allocated to either the training or testing set, thereby maintaining the integrity of the mutually exclusive division.

The real-world datasets

To further evaluate the clinical applicability of the models, we collected 5,582 images from 658 patients in 5 real-world datasets from 4 hospitals in China (Extended Data Fig. 2b). Specifically, 2,156 CT images from 171 patients diagnosed with stomach cancer and 1,516 CT images from 214 patients with colon cancer were incorporated from the Beijing Cancer Hospital. On the basis of the above 2 datasets, we evaluated the models on stomach tumour and colon tumour segmentation. In addition, 770 CT images from 53 pneumonia patients from Sun Yat-sen Memorial Hospital were incorporated to evaluate MedSegX's performance on 4 tasks, including consolidation, ground-glass opacity, interstitial thickening and pleural effusion. Futhermore, 371 ultrasound images from 20 patients with liver mass were incorporated from The Third Affiliated Hospital of Sun Yat-Sen University, enabling the evaluation of the model's clinical applicability on 2 specific tasks: hemangioma and hepatocellular carcinoma. Finally, 769 CT images from 200 patients with non-small-cell lung cancer from the West China Hospital of Sichuan University were incorporated to evaluate the models' performance on lung tumour segmentation. All cases in the real-world sets were split randomly into mutually exclusive sets for fine-tuning and testing at a ratio of 8:2.

MedSegX is a generalist medical segmentation model

We adopted SAM[15] as the backbone of our MedSegX, which consists of 3 main components: a context-guided image encoder, a prompt encoder and a mask decoder.

In this study, we integrated a Contextual Mixture of Adapter Experts (ConMoAE), consisting of a Hierarchical Structure-Based Contextual Embedding Prior (HScEP) and a Context-Guided Mixture of Adapters Network (CMoAN), into the SAM architecture (by integrating ConMoAE with the Feed Forward Network (FFN) of each Transformer layer). Specifically, HScEP maps the explicit knowledge priors from MedSegHierarchy into the corresponding contextual embedding as follows:

$$E = (e_1, e_2, e_3, e_4, e_5) \tag{1}$$

As Fig. 1c shows, e_1 corresponds to the top-level anatomical embedded prior denoting 'human body'. e_2, e_3 correspond to the sublevel anatomical embedded prior denoting 'body parts' and 'body subregions', respectively. e_4 is the embedded prior of the specific 'organs and tissues', and e_5 corresponds to the embedded prior of the 'task'. As considering a single tissue may involve multiple modalities, we additionally introduced modality embedding nodes M to provide explicit modality priors for MedSegX. Each embedding node in E and M has a corresponding learnable parameter vector with a length of D_e.

Then, a given image representation associated with a specific contextual embedding is received by the proposed CMoAN, which is composed of multiple adapter experts and a gating network assigning weight contributions to each adapter, allowing for dynamic data allocation among them. Each adapter was designed to act as a bottleneck structure with the dimension of D_a, consisting of a down-projection layer with parameters W_{down}, an up-projection layer with parameters W_{up}, and a nonlinear layer (ReLU[52]). Given an image representation R_i, the adapted feature $\widetilde{R}_i$ can be formalized by $\widetilde{R}_i = \text{ReLU}(R_i W_{\text{down}}) W_{\text{up}}$. Receiving the extracted contextual embedding as a condition, CMoAN can provide task-adaptive models by the flexible activation of a linear combination of different adapters. Specifically, in each Transformer layer, image representation R_i is concatenated with the embedded prior E_i and M_i to guide the gating network to determine the contribution ratio of each adapter for the ith data sample. We denote the number of adapters by K and introduce a gate vector $G \in \mathbb{R}^K$. Given any image representation R_i, the contextual embedding E_i and the modality embedding M_i, the gate vector is calculated as,

$$G = \text{softmax}(g(R_i \oplus E_i \oplus M_i)) \tag{2}$$

where $\oplus$ denotes a concatenation operator and g represents a gating network composed of a linear layer, which outputs K scores to assign the weights for each adapter. Then, the final result Y_i of CMoAN can be computed as follows,

$$Y_i = \text{FFN}(R_i) + \sum_{k=1}^{K} G_k f_k(R_i) \tag{3}$$

where FFN represents the feed forward network, f_k denotes the kth adapter, and G_k, the k th entry in G, indicates the contribution of the kth adapter to Y_i.

For the selection of prompts, we only focused on the bounding box prompt for simplicity. After obtaining the image embedding and the prompt embedding, the mask decoder mapped them to a segmentation mask using the Mask2Former[53]. We followed SAM[15] to reduce task ambiguity by predicting three masks during training. The loss between the ground truth and each of the predicted masks was calculated, but only the lowest loss was backpropagated. The mask prediction was supervised by a linear combination of cross-entropy loss and Dice loss[54] at a ratio of 1:1.

MedSegX3D to scale MedSegX to 3D medical segmentation

Furthermore, we extended the evaluation of our model to include 3D segmentation, specifically extending our model to MedSegX3D for volumetric segmentation. To extract the depth-dimensional information from 3D medical data, we replaced 2D adapters in CMoAN with 3D adapters. Each 3D adapter is composed of a down-projection layer with parameters W_{down}, a 3D convolutional layer Conv3D, an up-projection layer with parameters W_{up} and a nonlinear layer (ReLU). Given an image representation R_i, the adapted feature $\widetilde{R}_i$ can be formalized by $\widetilde{R}_i = \text{ReLU}(\text{Conv3}D(R_i W_{\text{down}})) W_{\text{up}}$. Specifically, given a batch of input 3D volumes V passed into the MedSegX3D, following MA-SAM[55], we first merged the adjacent slices of the 3D medical images into the batch dimension, thereby transforming $V \in \mathbb{R}^{B \times N \times C \times H \times W}$ into $V' \in \mathbb{R}^{BN \times C \times H \times W}$, where B denotes the batch size, N represents the number of slices, C denotes the channels of the slice, H is the height of the slice and W is the width of the slice. After converting images into patch embeddings, we reshaped the feature maps from $R \in \mathbb{R}^{BN \times C' \times H' \times W'}$ to $R' \in \mathbb{R}^{B \times N \times C' \times H' \times W'}$ before feeding them into the 3D convolutional layers of the adapter, where C' refers to the channels of the feature map, H' is the height and W' is the width, respectively. In this way, our MedSegX3D can effectively extract the volumetric information from 3D medical imaging.

Organized and unorganized tasks in MedSegHierarchy

While MedSegX relies on explicit knowledge priors extracted from MedSegHierarchy for inference, it still retains the capability to be applied to tasks that have not been incorporated into the hierarchy. For tasks already organized within MedSegHierarchy, MedSegX directly harnesses task-related entity nodes provided by the hierarchy, leveraging HScEP to map these priors into the embedding $E = (e_1, e_2, e_3, e_4, e_5)$ to facilitate both zero-shot inference and task-specific fine-tuning. For tasks not explicitly categorized within MedSegHierarchy, MedSegX utilizes HScEP to map relevant nodes from the anatomy layers (human body, body parts, body subregions, organs and tissues) of the hierarchy into embedded priors(e_1, e_2, e_3, e_4). Furthermore, it incorporates a fully zeroed vector e_* with a length of D_e to represent the relevant node from the 'task' layer, thereby obtaining a contextual embedding as $E = (e_1, e_2, e_3, e_4, e_*)$, enabling zero-shot inference and task-specific fine-tuning. This methodology enhances MedSegX's adaptability to tasks absent within the hierarchy.

Training and fine-tuning details

During training, we followed ref. 56 to determine hyperparameters by subdividing the initial training set at an 8:2 ratio, thereby obtaining 64% custom training and 16% validation sets. This validation set was first employed to determine the optimal hyperparameter configuration. Once the hyperparameters were established, we retrained the model on the union of custom train and validation sets, thus maximizing the use of all available medical data. In addition, AdamW optimizer with a learning rate of 0.001 and weight decay of 0.01 was utilized, supplemented by the cosine annealing learning rate scheduler. We trained MedSegX for 30 epochs on eight 80 GB NVIDIA A100 GPUs. The batch size was 1,024, and each run took ~120 GPU hours. After obtaining the trained MedSegX, the model was further fine-tuned on various cross-site and cross-target tasks to evaluate the generalization of the model. The most suitable learning rate was used for this purpose: $\{5\times10^{-6}, 1\times10^{-5}, 5\times10^{-5}, 1\times10^{-4}, 5\times10^{-4}\}$ The study initialized MedSegX with the pre-trained SAM-Base (94 M) model. The original SAM uses a Visual Transformer (ViT)[57] as an image encoder to handle high-resolution inputs with the size of $1,024 \times 1,024$. Considering the expensive training cost for input images with high resolution, we conducted the main experiments of MedSegX on the basis of input images with a resolution of 256×256. To maintain the output resolution with the size of 256, we added two additional deconvolution layers in the mask decoder, thus reducing training expenses. The length of the learnable parameter vector for embeddings D_e was set to 16. The number of adapters K was set to 4 and the bottleneck dimension of each adapter D_a was set to 16.

Comparison to other baselines

In this study, we compared our model against 5 baselines: 3 small models including (1) a task-specific nnU-Net[9], (2) a task-specific U-Mamba[25] and (3) a generalized DoDNet[13], as well as 2 foundation models including (1) MedSAM[18] and (2) SAM-Med2D[19]. For the task-specific nnU-Net and U-Mamba, we trained the model and then conducted the corresponding ID evaluation separately for each task of each ID dataset. For evaluating OOD performance, we examined a transfer learning setup between the ID and OOD sets. Specifically, for cross-site shifts, we transferred the pre-trained weights of nnU-Net on the same task from the ID set to the cross-site OOD set. For cross-task shifts, we conducted evaluation in 3 steps: First, we filtered out all ID tasks involving the same organ as the unseen target, on the basis of our MedSegHierarchy. Then, the pre-trained weights of nnU-Net or U-Mamba on the filtered ID tasks were transferred one by one to the unseen task, thereby performing zero-shot inference. Finally, we chose the best performance as the final zero-shot result and conducted the data-efficiency evaluation on the corresponding pre-trained model.

DoDNet utilizes a dynamic head for multitask segmentation. We trained DoDNet on large-scale mixed data and evaluated its ID performance using the task-related segmentation head for each task of each ID dataset. For evaluating OOD data, we utilized segmentation heads trained on the closest tasks from either the same or different ID datasets to perform zero-shot inference. Then, the best segmentation head for zero-shot performance was chosen for the data-efficiency evaluation.

In terms of large models, both MedSAM and SAM-Med2D use the same SAM-Base (94 M) backbone as used in our MedSegX. All segmentation tasks share the same set of parameters within the model. Specifically, MedSAM updates the parameters in the image encoder and mask decoder while keeping the parameters in the prompt encoder unchanged. SAM-Med2D inserts a general adapter into each Transformer layer of the image encoder and simultaneously fine-tunes the adapters, prompt encoder and mask decoder. These two baselines were both trained on mixed large-scale training data and then evaluated on each task of each ID and OOD dataset.

Moreover, we included WESAM[58] and SAM-COBOT[59] as two comparison models to test generalizability. To further validate the models' effectiveness on 3D segmentation, 5 SOTA methods, including 2 task-specific models (such as nnFormer[33] and MISSU[34]) and 3 foundation models (such as SAM-Med3D, Medical SAM 2 (ref. 35) and MedSAM2 (ref. 36)), were also incorporated.

Evaluation metrics and statistical analysis

All image segmentation tasks were evaluated using the Dice score and the 95% Hausdorff Distance (HD95), two widely used metrics for medical segmentation assessment. The Dice score measures the similarity between two sets by quantifying the overlap between the segmentation result and the ground truth:

$$\mathrm{DSC} = \frac{2TP}{2TP + FP + FN} \quad (4)$$

where TP, FP and FN refer to true positive predictions, false positive predictions and false negative predictions, respectively. The Dice score ranges from 0 to 1, where 1 indicates perfect overlap and 0 indicates no overlap. Therefore, a higher Dice score indicates better segmentation performance. HD95 was used to calculate the distance difference of boundaries between the segmentation result A and the ground truth B:

$$\mathrm{HD}_{95} = \max\left(h_{95}(A,B), h_{95}(B,A)\right) \quad (5)$$

$$h_{95}(A,B) = P_{95}(\{\min_{b\in B} d\left(a,b\right) | a \in A\}) \quad (6)$$

where $d\left(a,b\right)$ is the distance between points $a \in A$ and $b \in B$, and $P_{95}\left(\bullet\right)$ denotes the 95th percentile of the set of distances. In addition, we also calculated 95% CIs of the results for each task to assess the stability and confidence level of statistical results. For the overall performance, we took both instance- and task-level calculation into consideration, and all average results are presented as mean and standard deviation combined with 95% CIs. To compare our MedSegX with the most competitive baseline model, we conducted two-sided t-tests to determine whether significant differences exist. The statistical results were considered significant when the P value is less than 0.05.

Software materials

The study was implemented using Pytorch (2.0.0) on Python (3.10.12). For data analysis and results visualization, we also exploited several Python packages, including matplotlib (3.7.2), SimpleITK (2.2.1), numpy (1.25.2), opencv-python (4.9.0.80), pandas (2.1.4), Pillow (9.4.0), scikit-image (0.21.0), scipy (1.11.2), torchvision (0.15.0) and seaborn (0.13.0).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The main data supporting the results in this study are available within the paper and its Supplementary Information. The pre-training, ID validation and OOD validation datasets from our MedSegDB database are curated from open-source datasets and can be accessed via the weblinks provided in Supplementary Table 1. Among these, the data in MedSegDB that permit redistribution are available on HuggingFace[60] (https://huggingface.co/datasets/medicalai/MedSegDB). Each dataset used in the study is also provided in Supplementary Table 1. The validation dataset from MedSegDB that consists of real-world data collected from hospitals cannot be fully released in a public repository due to privacy regulations. We have deposited a minimum dataset of de-identified real-world data on GitHub (https://github.com/MedSegX/MedSegX-code).

Code availability

The source code to train MedSegX and reproduce the results is available in GitHub at https://github.com/MedSegX/MedSegX-code (ref. 61).

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Author contributions

G.W., X.L., Siqi Zhang, Q.Z., J. Yue, H.X., J. Yao and Y.W. collected and analysed the data. M.L., Shanghang Zhang and G.W. conceived and supervised the project. G.W., Siqi Zhang, X.L., Q.Z., Shanghang Zhang, J. Yue and M.L. contributed to data interpretation, method design, and critical review and wrote the manuscript. All authors discussed the results and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41551-025-01497-3.

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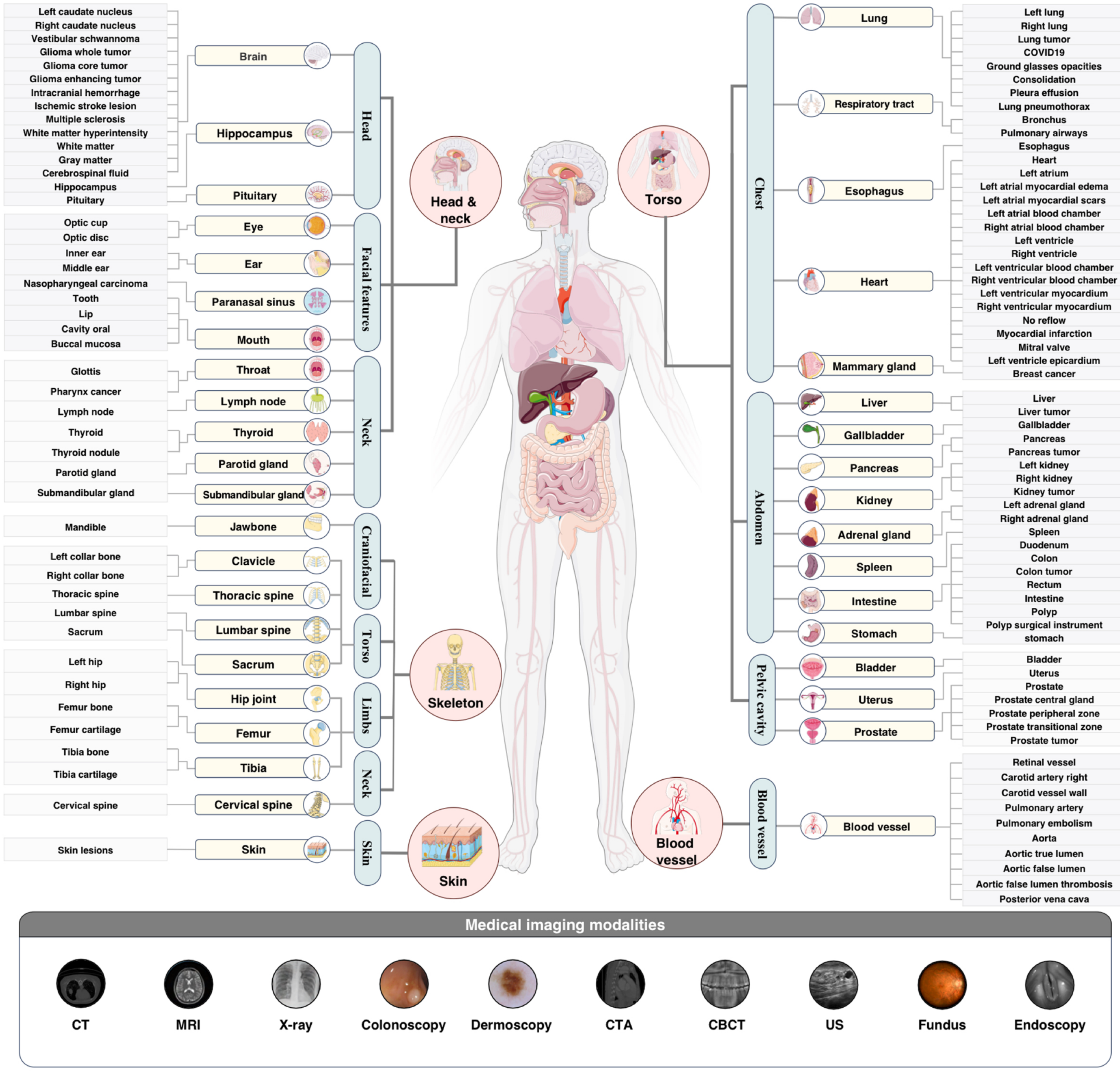

Extended Data Fig. 1 | Detailed description of MedSegDB. It consists of 5 anatomical subtrees, including the body parts of the *head and neck*, *torso*, *skeleton*, *blood vessel*, and *skin*, covering 39 major organs and tissues, and 111 segmentation tasks. MedSegDB incorporates 10 medical imaging modalities including the computed tomography (CT), magnetic resonance imaging (MRI), X-ray, colonoscopy, dermoscopy, CT angiography (CTA), cone beam CT (CBCT), ultrasound (US), fundus, and endoscopy.

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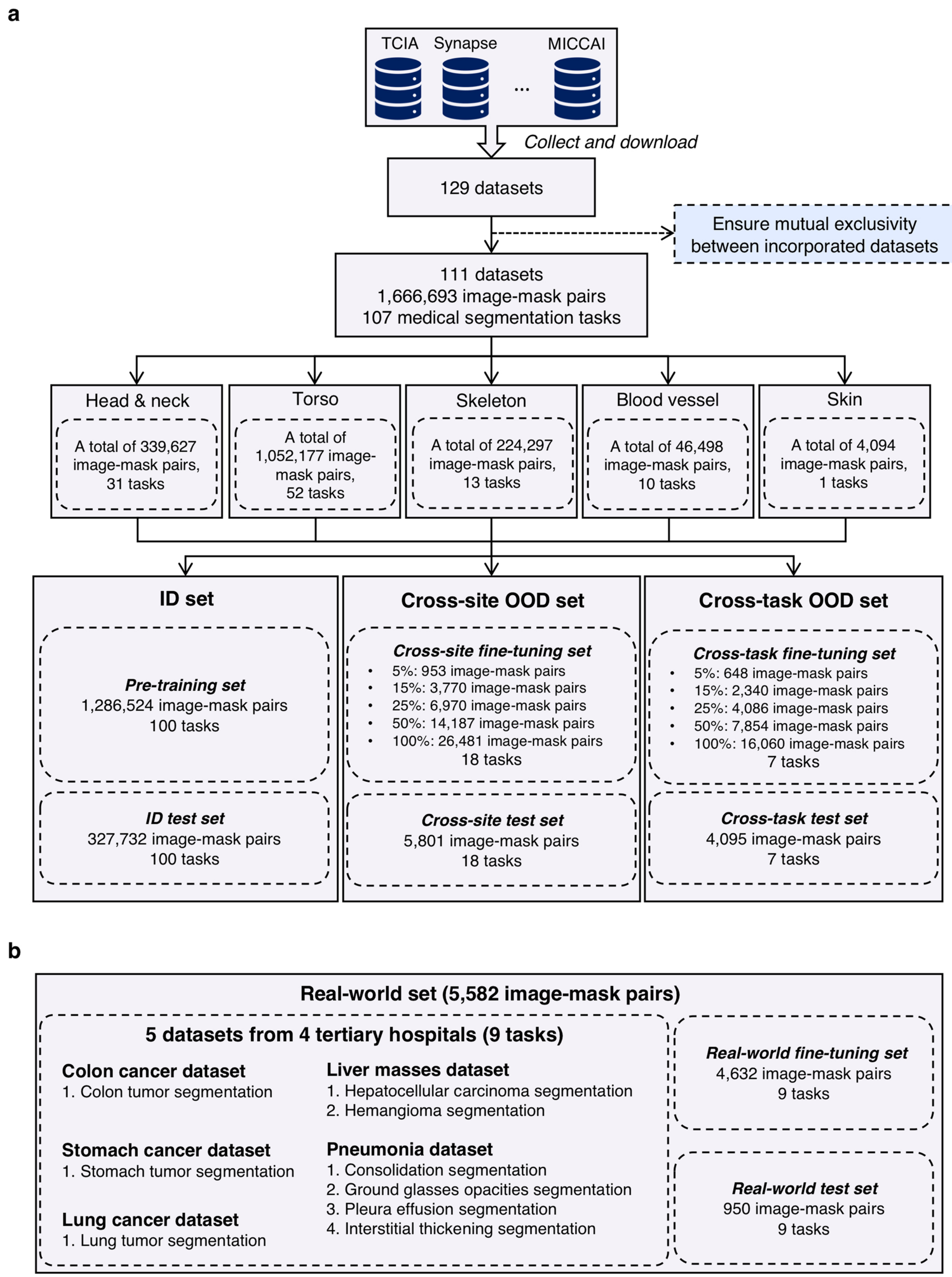

Extended Data Fig. 2 | Detailed data processing for development and evaluation of MedSegDB. a. Flow chart describing the datasets collected for the development and evaluation of ID and OOD sets. **b.** Details of the real-world datasets.

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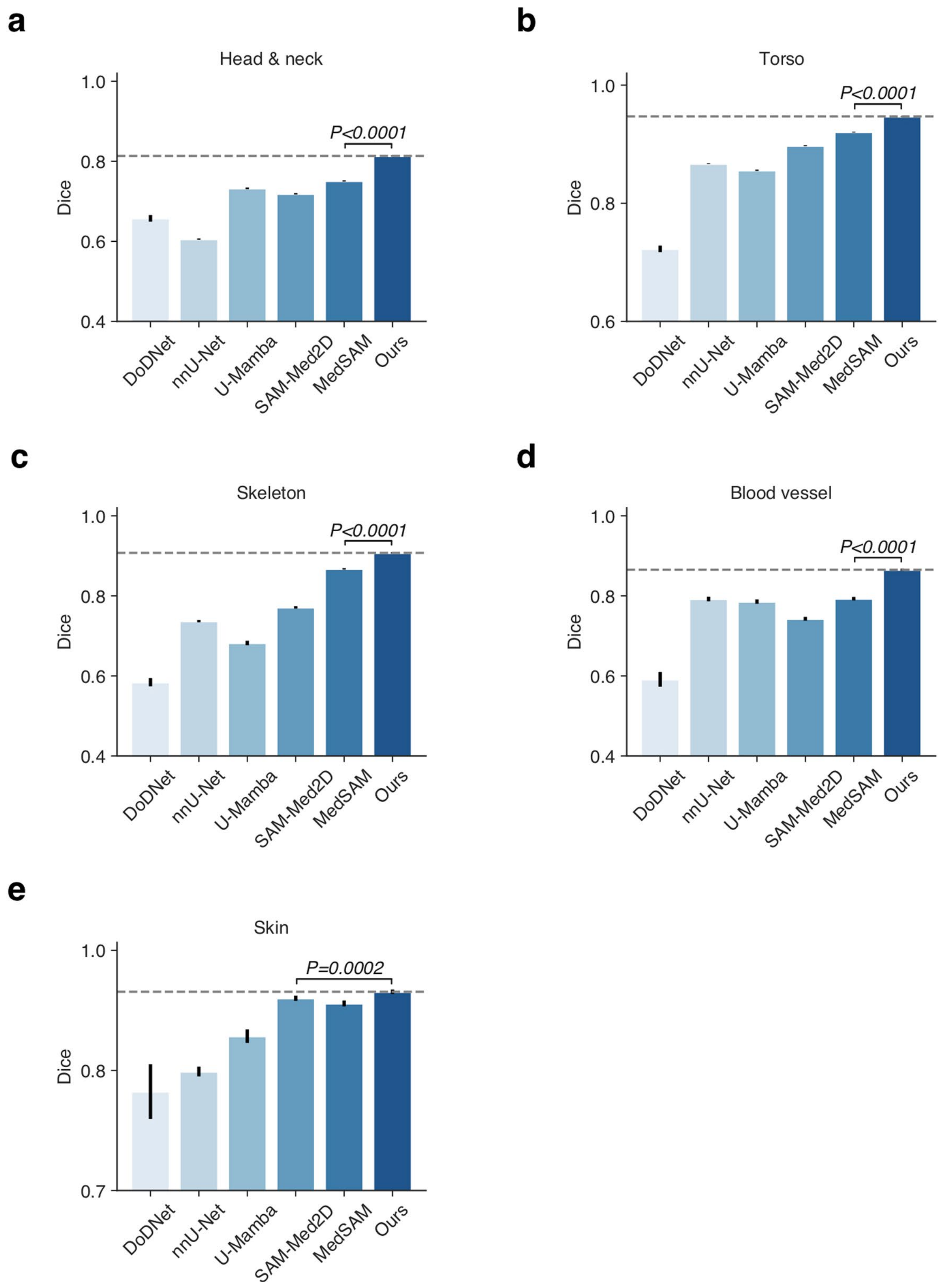
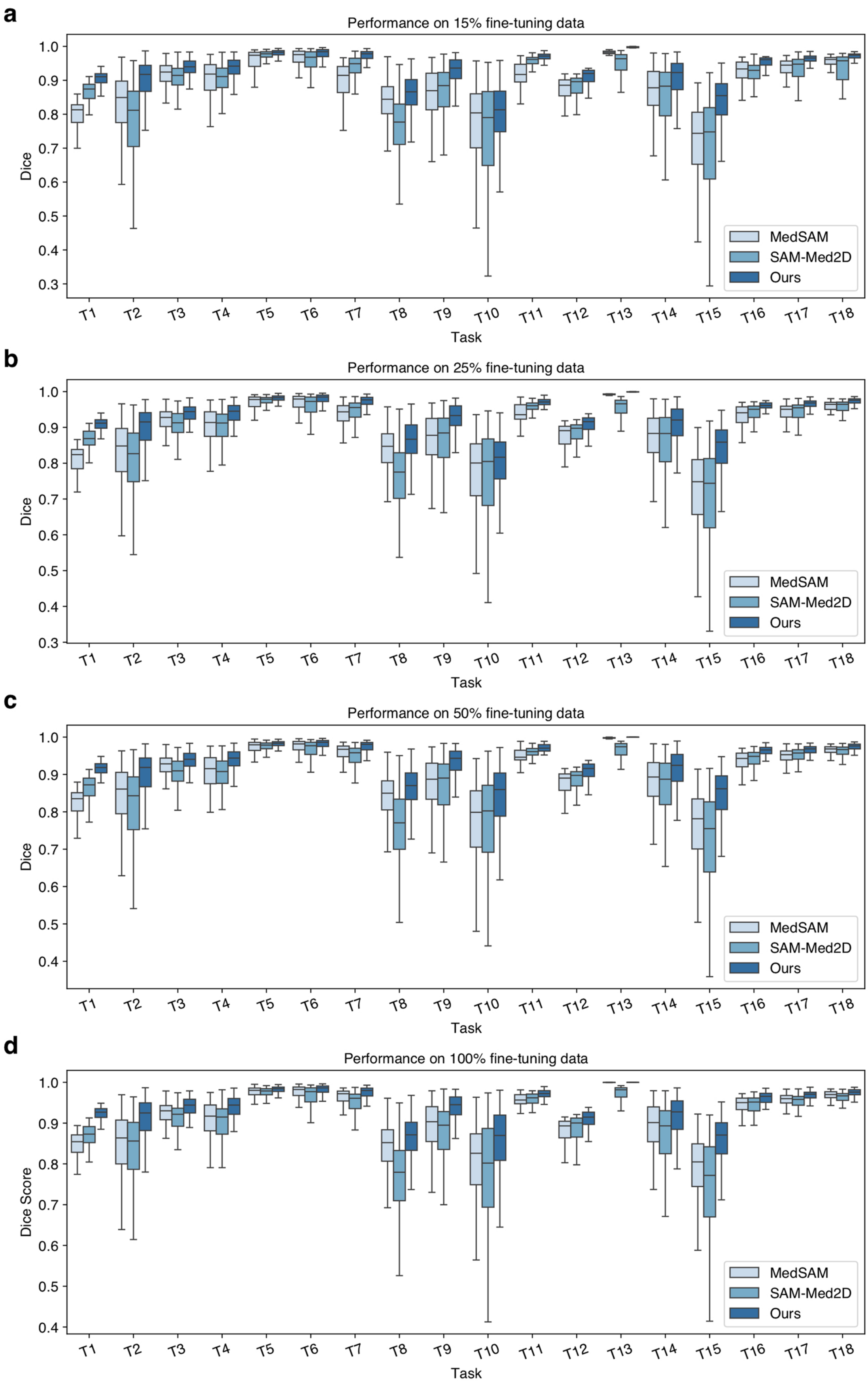


Extended Data Fig. 3 | Detailed performance of MedSegX and other competitors on tasks related to 5 body parts in ID test set. Comparison of 5 models on tasks in the parts of the **a**. *head and neck* (n = 67,746), **b**. *torso* (n = 205,066), **c**. *skeleton* (n = 45,981) **d**. *blood vessel* (n = 8,121), and **e**. *skin* (n = 818). P-values are calculated with two-sided t-test. Bar graphs indicate the mean ± 95% CI. The dashed horizontal line represents the Dice score achieved by MedSegX.



Extended Data Fig. 4 | See next page for caption.

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Extended Data Fig. 4 | Generalization performance evaluation of MedSegX and other competitors on 18 cross-site tasks with different proportions of OOD fine-tuning data (n = 5,801). a. 15% fine-tuning data. **b**. 25% fine-tuning data. **c**. 50% fine-tuning data. **d**.100% fine-tuning data. Tasks consist of tooth (X-ray), gallbladder (CT), left kidney (CT), right kidney (CT), left lung (CT), right lung (CT), liver (CT), pancreas (CT), stomach (CT), optic cup (Fundus), optic disc (Fundus), left atrium (MRI), prostate (MRI), right ventricle (MRI), left ventricle (MRI), spleen (MRI), left lung (X-ray), and right lung (X-ray) segmentation. In all box plots, each box shows the quartiles of the distribution, with center as the median, minimum as the first quartile, and maximum as the third quartile. The whiskers extend to the farthest data point that lies within 2× interquartile range (IQR) from the nearest quartile.

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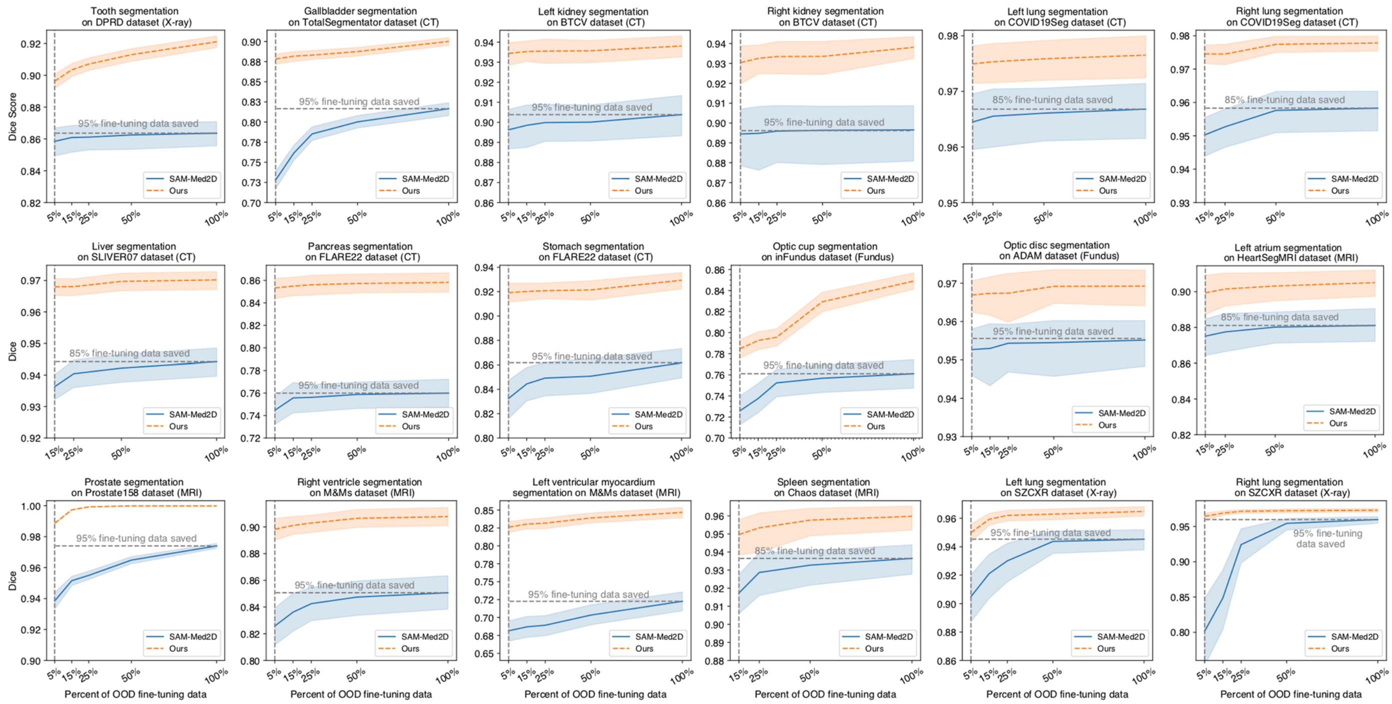

Extended Data Fig. 5 | Data-efficient performance evaluation of MedSegX and SAM-Med2D on 18 cross-site tasks. 95% CI intervals are shown by the shaded area.

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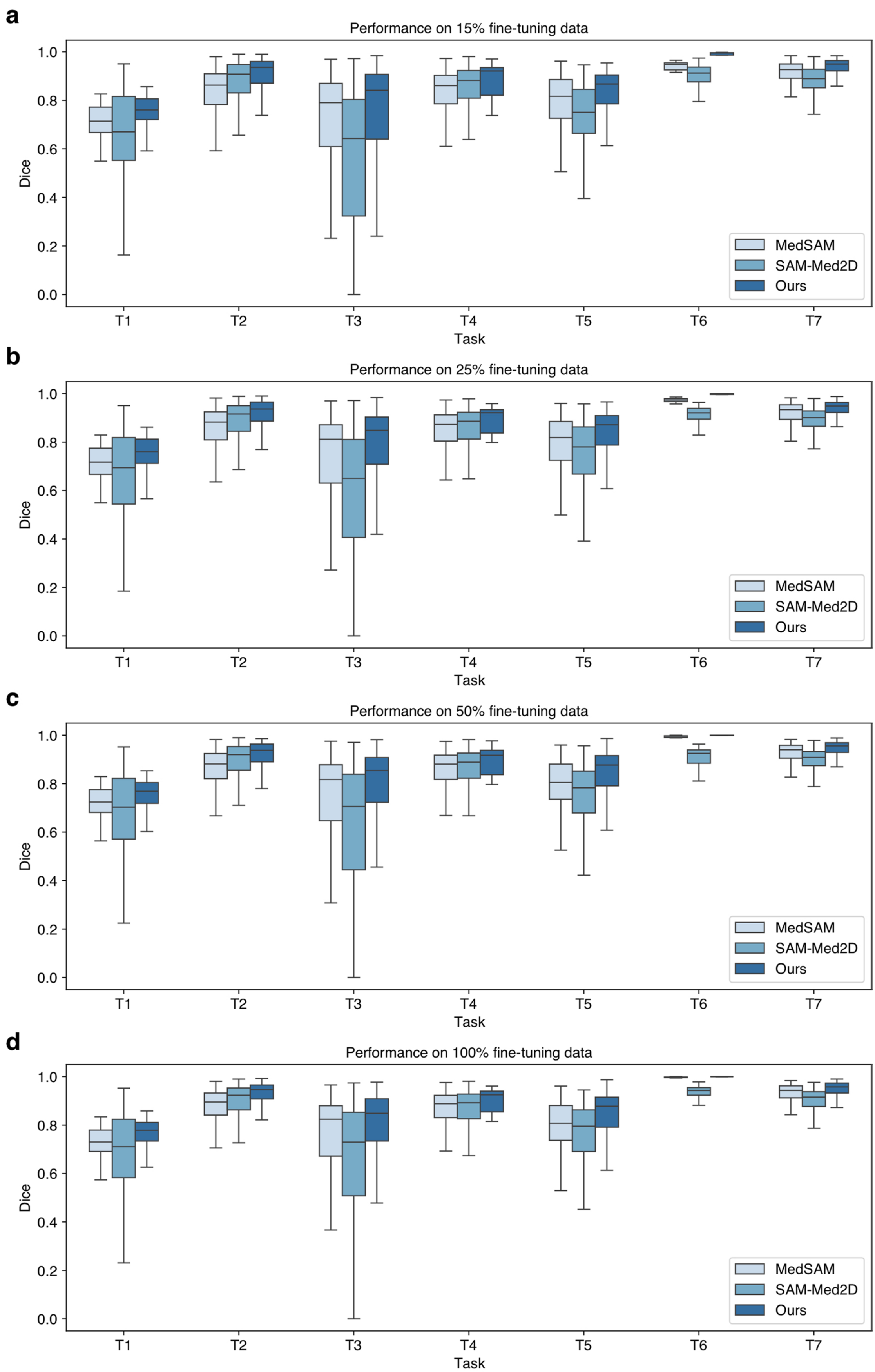

Extended Data Fig. 6 | See next page for caption.

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Extended Data Fig. 6 | Generalization performance evaluation of MedSegX and competitors on seven cross-target tasks with different proportions of OOD fine-tuning data (*n* = 4,095). a. 15% fine-tuning data. **b**. 25% fine-tuning data. **c**. 50% fine-tuning data. **d**.100% fine-tuning data. Tasks consist of the colon tumor (CT), kidney tumor (CT), liver tumor (CT), lung tumor (CT), pancreas tumor (CT), prostate tumor (MRI), and vestibular schwannoma segmentation (MRI). In all box plots, each box shows the quartiles of the distribution, with center as the median, minimum as the first quartile, and maximum as the third quartile. The whiskers extend to the farthest data point that lies within 2× interquartile range (IQR) from the nearest quartile.

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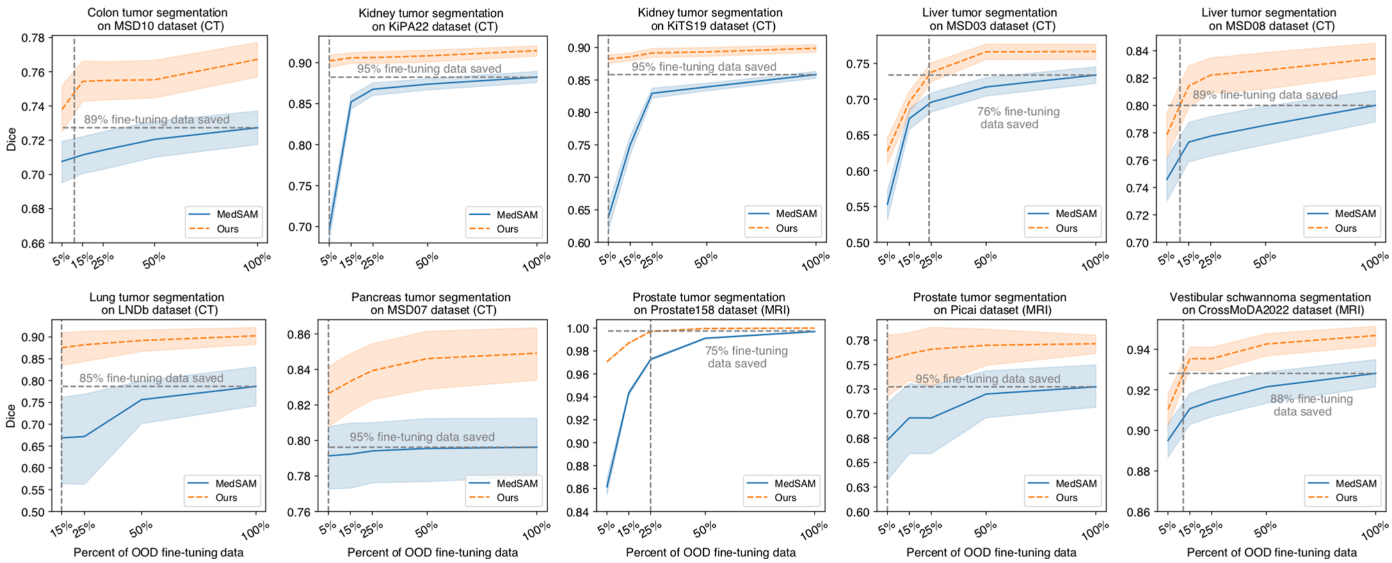

Extended Data Fig. 7 | Data-efficient performance evaluation of MedSegX and MedSAM on 7 unseen categories of body tumors. 95% CI intervals are shown by the shaded area.

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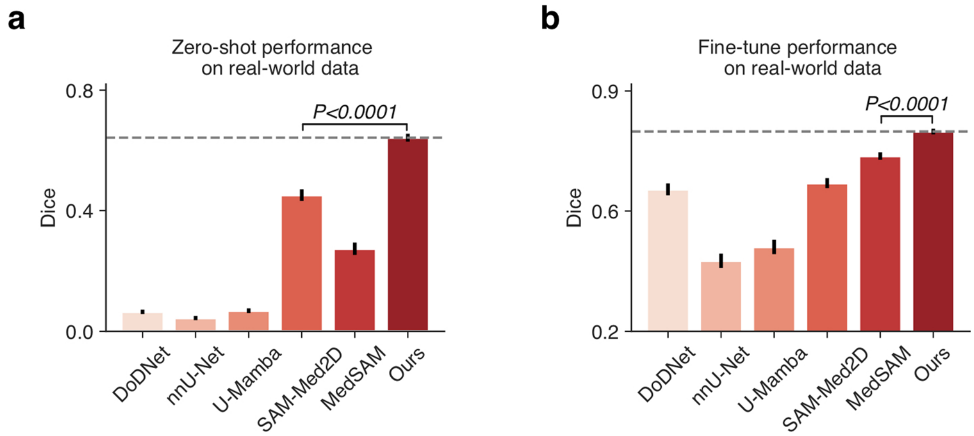

Extended Data Fig. 8 | The models' average performance under real-world setting (n = 950). a. Average zero-shot performance evaluation of 6 models on 5 real-world datasets. **b**. Average fine-tuning performance evaluation of 6 models on 5 real-world datasets. P-values are calculated with two-sided t-test. Bar graphs indicate the mean ± 95% CI. The dashed horizontal line represents the Dice score achieved by MedSegX.

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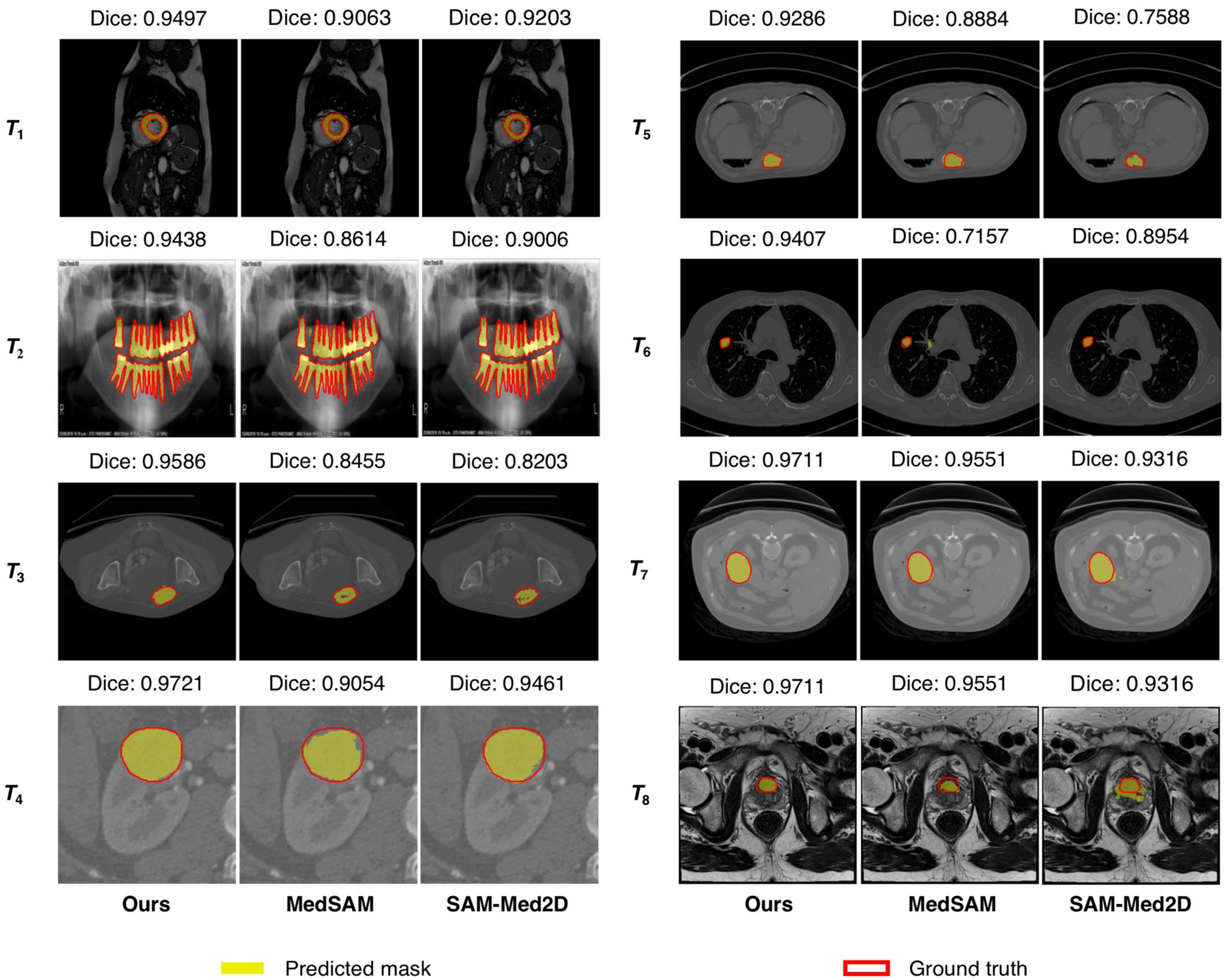

Extended Data Fig. 9 | Visualization of 8 segmentation examples in OOD test sets. T_1: Left Ventricular Myocardium, T_2: Teeth, T_3: Colon Tumor, T_4: Kidney Tumor, T_5: Liver Tumor, T_6: Lung Tumor, T_7: Pancreas Tumor, T_8: Prostate Tumor.

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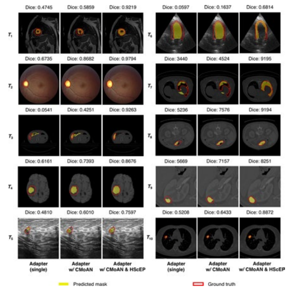

Extended Data Fig. 10 | Visualization analysis for the ablation experiments. 8 segmentation examples from both ID and OOD test sets are incorporated (ID tasks: T_1 ~ T_4. OOD tasks: T_5 ~ T_8): T_1: Left Ventricular Myocardium, T_2: Optic Disc, T_3: Liver, T_4: Brain Core Tumor, T_5: Breast Cancer, T_6: Left Ventricle Epicardium, T_7: Left Lung, T_8: Stomach, T_9: Kidney Tumor, T_{10}: Lung Tumor.

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Reporting Summary

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Scripts for data collection and preparation were written in Python (3.10.12), Pandas (2.1.4), numpy (1.25.2), and SimpleITK (2.2.1).
Data analysis	The study is implemented using Pytorch (2.0.0) on Python (3.10.12). As for data analysis and result visualization, we also exploit several Python packages, including matplotlib (3.7.2), SimpleITK (2.2.1), numpy (1.25.2), opencv-python (4.9.0.80), pandas (2.1.4), Pillow (9.4.0), scikit-image (0.21.0), scipy (1.11.2), torchvision (0.15.0) and seaborn (0.13.0). The codes are available for scientific research and non-commercial use on https://github.com/MedSegX/MedSegX-code.

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The main data supporting the results in this study are available within the paper and its Supplementary Information. The pre-training, in-distribution (ID) validation, and out-of-distribution (OOD) validation datasets from our MedSegDB database are curated from open-source datasets and can be accessed via the weblinks provided in Supplementary Table 1. Among those, the data in MedSegDB that permit redistribution are available on HuggingFace60 (https://huggingface.co/datasets/medicalai/MedSegDB). Each dataset used in the study has also been cited in the Supplementary Table 1. Regarding the validation dataset from MedSegDB that consists of real-world data collected from hospitals, due to privacy regulations, these data cannot be fully released in a public repository. We have deposited a minimum dataset of de-identified real-world data on GitHub (https://github.com/MedSegX/MedSegX-code).

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	We did not perform sex and gender analysis.
Reporting on race, ethnicity, or other socially relevant groupings	We did not collect or use any covariates regarding race, ethnicity, and other social groupings at any stage of the study.
Population characteristics	We did not collect or use any covariates pertaining to population characteristics at any stage of the study.
Recruitment	No patient recruitment was necessary in this study.
Ethics oversight	Ethical approval for this study was obtained from the institutional ethics committees with the approve number of 2020 (232). Only de-identified retrospective data was used for research, without the active involvement of patients.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	We collected 1,672,275 image-mask pairs from 134 medical segmentation datasets, covering ten medical imaging modalities, 39 major organs and tissues, and 111 segmentation tasks. Sample size was determined by the data availability. No additional statistical methods for sample size estimation were used.
Data exclusions	To ensure the quality of collected medical data, we excluded data samples with incomplete images or missing labels.
Replication	Our constructed database and corresponding codes are publicly available for replication.
Randomization	The datasets were randomly divided into development and evaluation sets without patient-level overlapping.
Blinding	During the data processing, all data were first de-identified to remove any patient related information.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study	
☒	☐	Antibodies
☒	☐	Eukaryotic cell lines
☒	☐	Palaeontology and archaeology
☒	☐	Animals and other organisms
☒	☐	Clinical data
☒	☐	Dual use research of concern
☒	☐	Plants

Methods

n/a	Involved in the study	
☒	☐	ChIP-seq
☒	☐	Flow cytometry
☒	☐	MRI-based neuroimaging

Plants

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.